IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 17 June 2025, accepted 1 July 2025, date of publication 10 July 2025, date of current version 16 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3587345

SURVEY

Blockchain-Enabled Federated Learning in Healthcare: Survey and State-of-the-Art

NASIM NEZHADSISTANI[1,2], (Graduate Student Member, IEEE), NAGHMEH S. MOAYEDIAN[1], (Member, IEEE), AND BURKHARD STILLER[2], (Member, IEEE)

[1]Department of Electrical and Computer Engineering, Isfahan University of Technology, Isfahan 8415683111, Iran
[2]Communication Systems Group (CSG), Department of Informatics (IfI), University of Zürich (UZH), 8050 Zürich, Switzerland

Corresponding author: Nasim Nezhadsistani (nezhadsistani@ifi.uzh.ch)

This work was supported partially by (a) the University of Zürich (UZH), Switzerland, and (b) the Horizon Europe Framework Program's project Net-Zero self-adaptive activation of distributed self-resilient augmented services (NATWORK), Grant Agreement No. 101139285, funded by the Swiss State Secretariat for Education, Research, and Innovation SERI, under Contract No. 23.00642.

ABSTRACT Advances in Internet of Medical Things technology, information and communication technologies, and machine learning have initiated the shift in healthcare towards smart healthcare. Centralization of health data to train ML models does pose privacy, ownership, and regulatory problems. Federated learning solves such problems by distributing the learning process to several devices, but it also encounters problems like encouraging participants and model aggregation correctness. Combining blockchain and FL can solve such problems through a decentralized approach that provides greater security and privacy for intelligent healthcare. This survey provides a systematic review of blockchain-based federated learning (BCFL) systems in healthcare. Key design features of BCFLs are analyzed, such as consensus protocols, crypto protocols, storage topology, and integration processes relevant to healthcare use cases. Characteristics such as convergence delay, computation overhead, accuracy loss when privacy is an issue, and ledger scalability for different implementations are compared among common implementations. The works of recent FL-based healthcare frameworks have been discussed, along with determining the challenges and research directions for healthcare use cases.

INDEX TERMS Blockchain, data sharing, EHR, federated learning, IoMT, privacy.

I. INTRODUCTION

In the landscape of healthcare technology, an array of sensors and wearable devices perpetually collect data, forming a vast reservoir of medical information [1]. Traditionally, these data have been consolidated in centralized storage systems, but concerns regarding security vulnerabilities and privacy breaches have spurred the exploration of alternative approaches. Federated learning (FL) has emerged as a promising solution, enabling machine learning algorithms to be trained on decentralized data stored across various devices, thus preserving data privacy while still facilitating model training [2], [3]. FL not only addresses concerns about data security but also enhances operational efficiency by reducing latency and providing personalized recommendations. However, challenges, such as potential server failures and malicious interventions, persist. FL enables hospitals and healthcare providers to store sensitive patient information locally while still contributing to a shared global model. This preserves data privacy and complies with regulations such as the Health Insurance Portability HIPAA and the General Data Protection Regulation (GDPR) [4], [5]. The global model is updated by aggregating the model parameters rather than raw data, thereby reducing the security risks associated with centralized data storage [6]. IoT medical devices and wearable sensors are increasingly used to collect real-time patient health data. FL allows this data to be utilized for training AI models without compromising privacy [7], [8]. This is particularly valuable given the sensitive nature of medical data and the legal restrictions on sharing it between

The associate editor coordinating the review of this manuscript and approving it for publication was Sedat Akleylek.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

TABLE 1. Summary of existing surveys.

Application	FL	BC	Technical Challenges	Research Direction	Summary	Ref
Privacy Preservation in Smart Healthcare	H	L	M	M	This paper examines how FL can enhance privacy in smart healthcare systems, particularly within IoMT networks, by employing techniques like differential privacy and reinforcement learning. It also surveys various FL algorithms and architectures, emphasizing their potential to safeguard medical data and advance secure, efficient healthcare systems.	[13]
Blockchain and ML in EHR Security	M	H	L	M	This research paper explores how combining blockchain technology with machine learning can enhance data security, interoperability, and patient privacy in healthcare. The study addresses challenges like data integrity, scalability, and privacy in EHR by using machine learning for disease prediction and treatment optimization and blockchain for secure data storage.	[14]
FL and its application in Healthcare	H	M	M	L	This paper reviews recent literature on federated learning in healthcare, explores various architectures and models, and examines the security risks and privacy protection methods. It concludes with a summary and a future outlook on the application and security of federated learning in healthcare.	[15]
Blockchain meets FL in Healthcare	H	H	M	L	This paper explores the integration of blockchain and FL for healthcare, highlighting benefits and limitations, and emphasizing three main research areas: IoMT, EHR/EMR management, and digital healthcare systems. It also presents a conceptual framework for blockchain-enabled FL.	[16]
Privacy-preserving with blockchain and FL for the Telemedicine	H	H	L	M	This study reviews literature on privacy-preserving methods using blockchain and federated learning in telemedicine, providing a qualitative analysis of architectures, privacy mechanisms, and machine learning methods for data storage, access, and analytics. The study aims to integrate these technologies to design a secure, trustworthy, and accurate telemedicine model with guaranteed privacy.	[17]
Blockchain-enabled FL in Healthcare	H	H	H	H	This survey gives end-to-end analyses of blockchain-based federated learning (BCFL) in the healthcare field, with major applications like EHR management, IoMT, diagnosis, precision medicine, outbreak prediction, and drug discovery. Unlike other surveys, it emphasizes system-level outcomes, such as implementation metrics, regulatory compliance (HIPAA, GDPR, EU AI Act), and crypto security (ZKPs, post-quantum signatures). It also identifies critical gaps in scalability, accuracy, and real-time guarantees, offering a structured research agenda to guide future BCFL development.	Our Survey
L Low Significance			M Medium Significance		H High Significance	

institutions [9]. However, challenges remain in implementing FL in healthcare. Medical data are often non-identically distributed across institutions because of the differences in patient populations. This can reduce the accuracy of global models [10]. Additionally, there are concerns regarding potential vulnerabilities in the central server aggregating model updates. With the increasing volume of medical data integral to patient care, it is crucial to safeguard against cyberattacks. Although external threats can be mitigated through authentication and encryption, internal threats, such as privacy breaches and denial-of-service attacks, pose significant risks. To address these issues, researchers have explored the integration of blockchain technology with FL. This can further decentralize the process and enhance security. Integration of blockchain technology with FL offers a potential solution by decentralizing data storage and ensuring data integrity through immutable ledgers [11]. This fusion presents a transformative opportunity in healthcare, maximizing the utilization of medical data while safeguarding patient privacy, thereby fostering advancements in research and enhancing Patient Care on a Global Scale. This approach could unlock new research and economic opportunities in the field while improving patient care globally [11], [12]. Table 1 summarizes a comparison of existing surveys on blockchain and FL integration in healthcare applications [13], [14], [15], [16], [17].

A. MOTIVATION AND OBJECTIVE

The rapid advancement of healthcare technologies, particularly through the proliferation of Internet of Medical Things (IoMT) devices and AI-driven diagnostic tools, has transformed the way medical data is generated and utilized. However, centralized machine learning models commonly used in clinical environments raise serious concerns related to patient privacy, data ownership, and regulatory compliance. These issues have prompted growing interest in FL, which enables decentralized model training without the need to share raw data. While FL improves privacy, it still depends on a central aggregator and lacks inherent mechanisms for auditability, integrity, and trust. Integrating blockchain technology with FL presents a promising solution: blockchain's decentralized and tamper-resistant architecture complements

FL by enhancing data provenance, trust, and system robustness.

Despite increasing interest in this integration, prior studies have often focused on either FL or blockchain in isolation, lacking a holistic analysis of their combined potential. Notably, critical aspects such as implementation-level trade-offs (e.g., gas consumption, confirmation latency), advanced cryptographic mechanisms (e.g., Secure Multi-Party Computation (SMPC), zero-knowledge proofs (ZKPs), post-quantum signatures), and regulatory compliance frameworks (HIPAA, GDPR, EU AI Act) remain underexplored in the literature.

This study is motivated by the urgent need to address these research gaps and systematically evaluate Blockchain-enabled Federated Learning (BCFL) as a foundation for secure, privacy-preserving, and regulation-aware healthcare systems. The objective is to assess how BCFL can support trustworthy data sharing, resilient model training, and decentralized decision-making in real-world healthcare environments. In doing so, this work also aims to identify the practical challenges of BCFL deployment and propose research directions to guide its future development.

B. CONTRIBUTION

This study makes substantial contributions to the convergence of healthcare, blockchain technology, and FL, highlighting the transformative potential of this integration for enhancing data privacy, security, and efficiency in healthcare applications. The key contributions of this study are as follows.

- **Comprehensive Survey of Blockchain FL in Healthcare**: This study provides a detailed review of the integration of blockchain and FL in healthcare, critically evaluating the benefits and challenges of these technologies. It explores how blockchain enhances the security and privacy of FL by addressing issues such as central server failures and unauthorized data manipulation, particularly in sensitive applications such as electronic health record (EHR) management and IoMT. This study highlights how blockchain can improve the reliability and resilience of decentralized healthcare systems.
- **Discussion of BCFL frameworks**: This study discusses existing conceptual frameworks and approaches for BCFL in healthcare. It reviews how these frameworks address the challenges related to secure and scalable data sharing, model validation, and maintaining patient confidentiality, emphasizing their potential to enhance collaboration across healthcare institutions.
- **Identification of Key Applications**: This study categorizes the practical applications of BCFL, including medical diagnostics, precision medicine, disease outbreak prediction, drug discovery, and healthcare resource allocation. Case studies have demonstrated how these technologies enhance the accuracy, reliability, and personalization of healthcare models.

TABLE 2. Database search strings and PRISMA screening summary.

Database / PRISMA Stage	Search String / Description	Records (n)
Search Results by Database		
IEEE Xplore	`("federated learning") AND ("blockchain") AND (health* OR medical)`	204
Springer Link	`blockchain AND federated learning AND (health* OR medical)`	247
ScienceDirect	`"blockchain" AND TITLE("federated learning") AND (health* OR medical)`	118
ACM Digital Library	`Title("federated learning") AND Title/Abs(blockchain) AND (health*)`	63
PubMed	`("federated learning"[tiab]) AND blockchain[tiab] AND (health* OR medical)`	58
Scopus	`TITLE-ABS-KEY("federated learning") AND ("blockchain") AND (health* OR medical)`	400
PRISMA 2020 Screening Workflow Summary		
Records identified from databases	Total before duplicate removal	1,090
Duplicate records removed	Automated + manual	220
Records screened (titles/abstracts)	After duplicates removed	870
Records excluded	Irrelevant titles / abstracts	600
Full-text articles assessed	Eligibility assessment	270
Full-text articles excluded	Did not meet criteria	120
Studies included in synthesis	Final eligible studies	**150**

- **Analysis of Technical Challenges and Solutions**: This study examines challenges such as data heterogeneity, communication overhead, and scalability in BCFL systems. It discusses existing solutions, such as SMPC, and proposes methods to address interoperability and aggregation issues.
- **Identification of Future Research Directions**: This study highlights key areas for future research on BCFL, focusing on challenges such as scalability, enhanced privacy techniques, interoperability, and regulatory compliance. These approaches aim to improve the performance, security, and practical deployment of these technologies for large-scale healthcare applications.

C. SURVEY METHODOLOGY

This review was created and reported according to the PRISMA 2020 guideline [18]. The process includes five steps: (i) database selection, (ii) query formulation, (iii) title/abstract screening, (iv) full-text eligibility evaluation, and (v) qualitative and quantitative synthesis.

1) LITERATURE SEARCH STRATEGY

A systematic search took place in six academic databases: *IEEE Xplore*, *Springer Link*, *ScienceDirect*, *ACM Digital Library*, *PubMed*, and *Scopus*. Boolean query terms were tailored for each database to find instances of "*federated learning*" and "*blockchain*" in healthcare contexts (e.g., `TITLE-ABS-KEY("federated learning") AND ALL("blockchain") AND (health OR medical)` for Scopus). Table 2 summarizes the searched exact strings and the number of records returned.

2) ELIGIBILITY, SCREENING, AND DATA EXTRACTION

Full-length articles, conference proceedings, and book chapters from peer-reviewed journals in English, published between 2018 and 2025, were included if they clearly integrated federated learning (or its closest decentralized variant) with blockchain at any step in healthcare data

collection, storage, analysis, or governance. Preprints, theses, tutorials, editorials, manuscripts in a language other than English, and investigations without an implemented or experimentally tested BCFL system were excluded. Database searching yielded 1090 records; 220 duplicates were removed, and the remaining 870 were original titles and abstracts. Blinded assessors screened the records separately with a pilot, achieving a Cohen's *kappa* of 0.89, and discarded 600 items as non-relevant. Full-text screening of the remaining 270 articles against the inclusion criteria and another 120 articles, producing a final synthetic set of 150 studies. For each paper from the considered ones, twelve analytical variables were harvested: clinical domain; FL paradigm (cross-device, cross-silo or vertical); learning architecture; blockchain platform and consensus protocol; privacy or fairness mechanisms; adversarial model accounted for; data modality; heterogeneity mitigation technique; scalability of properties; performance metrics reported; regulatory conformity and reproducibility; artifact availability (code or datasets). Each article was assigned a tailor-made *Gap-Score* (0–3) capturing the depth of implementation, cryptographic embedding, and regulatory awareness.

D. PAPER STRUCTURE

The rest of this paper is structured as follows. Section II provides a brief overview of blockchain and FL technologies, along with their respective characteristics. It also discusses how they can be combined to enhance healthcare data security and privacy. This section presents the conceptual building blocks to pursue the following analyses. Section III provides a detailed description of the integration of FL and blockchain, focusing on their privacy and security features for applications in smart healthcare. The technical aspects of blockchain, for instance, through the use of cryptographic mechanisms and consensus algorithms, are discussed in detail in this section. Section IV discusses some of the possible applications of BCFL in healthcare, for instance, medical condition diagnosis, precision medicine, disease outbreak prediction, and resource allocation. Section IV documents actual applications and considers the extent to which these technologies improve healthcare outcomes. Section V examines the challenges and potential future directions for applying BCFL to healthcare, including regulatory considerations, scalability, and technological limitations. Section VI summarizes the research results and emphasizes the importance of additional research in this field.

II. BACKGROUND

This section provides a comprehensive overview of blockchain and federated learning technologies, highlighting their benefits, limitations, and applications. In addition, it examines the integration of blockchain and federated learning.

A. BLOCKCHAIN

Blockchain is a decentralized system that revolutionizes data management by creating a secure and transparent digital

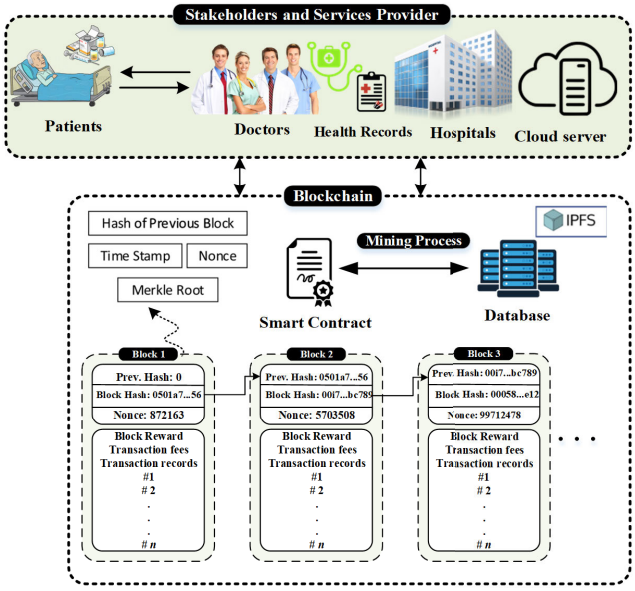


FIGURE 1. Blockchain in decentralized healthcare.

ledger. It consists of interconnected blocks that store transactional data in chronological order, ensuring data integrity and immutability [19]. As shown in Figure 1, each block contains a unique identifier called a hash, which is generated based on its content, making it tamper-proof. The chain structure of the blocks ensures that any alteration in one block is immediately detected by subsequent blocks, maintaining the integrity of the entire chain [20]. A key advantage of blockchain is its decentralized nature, which eliminates the need for intermediaries in transactions. This peer-to-peer network enables direct and secure data exchange between participants, thereby reducing costs and enhancing efficiency. In addition, blockchain enhances transparency by providing a shared view of data across all network participants, thereby promoting trust and accountability. The distributed consensus mechanism ensures that all network nodes agree on the validity of transactions, enhances security, and prevents fraud [21], [22]. Blockchain technology has applications across various industries, including finance, healthcare, and supply chains. In healthcare, blockchain ensures secure and interoperable exchange of EHR, improving patient care coordination and data accuracy. Supply chain management benefits from blockchain's ability to track and authenticate products throughout the supply chain, reducing counterfeiting and enhancing transparency [23], [24]. Financial institutions are leveraging blockchain technology for secure and efficient cross-border transactions, thereby eliminating intermediaries and reducing transaction costs. The integration of blockchain with other technologies, such as artificial intelligence and IoT, opens up new possibilities for innovation. Blockchain-powered smart contracts automate and secure digital agreements, reducing the need for manual intervention and streamlining processes [25]. The combination of blockchain and AI enables data-driven insights and predictive analytics,

enhancing decision-making and business intelligence. IoT devices connected to blockchain networks can securely exchange data and execute transactions autonomously, creating a seamless and secure ecosystem [26].

B. FEDERATED LEARNING

Federated learning is a machine learning approach that enables training models across decentralized devices or servers while keeping data localized. This privacy-preserving technique allows collaborative model training without sharing raw data, thereby addressing privacy concerns associated with centralized data storage. Each device or server computes model updates based on local data and shares updates only with a central server or aggregator, thereby ensuring data privacy and security [27]. A key advantage of federated learning is its ability to leverage data from multiple sources without compromising individual privacy. By training models locally on user devices or servers, federated learning enables personalized model updates while preserving data confidentiality. This distributed learning approach is particularly beneficial in scenarios where data cannot be easily centralized, owing to privacy regulations, data sensitivity, or network constraints [28]. Federated learning finds applications in various domains, including healthcare, finance, and IoT. In healthcare, federated learning enables collaborative model training on patient data from different healthcare providers without sharing sensitive information, thereby improving the accuracy of medical AI models [29].

In the financial sector, federated learning enables banks to collaborate on fraud detection models without sharing customer transaction data, thereby enhancing security and compliance [30]. The decentralized nature of federated learning also contributes to the scalability and efficiency of large-scale machine learning tasks. By distributing model training across multiple devices or servers, federated learning reduces the computational burden on the central server and accelerates the learning process. This distributed approach enables real-time model updates and continuous learning without the need for extensive data transfers, making it suitable for dynamic and resource-constrained environments [31], [32], [33]. The objective is to minimize the following global objective:

$$\mathcal{L}(\theta) = \frac{1}{\sum_{k=1}^{K} n_k} \sum_{k=1}^{K} n_k \, \ell_k(\theta), \tag{1}$$

K represents the number of clients, n_k represents the number of samples at each client, and $/ell_k(/theta)$ represents its empirical loss on its local dataset D_k. Weighting by n_k ensures that clients contribute proportionally to the dataset size, which is particularly significant in healthcare applications with imbalanced data distributions.

1) LOCAL TRAINING

For every communication round t, the global model $\theta^{(t)}$ is propagated to all the participating clients. Every client runs

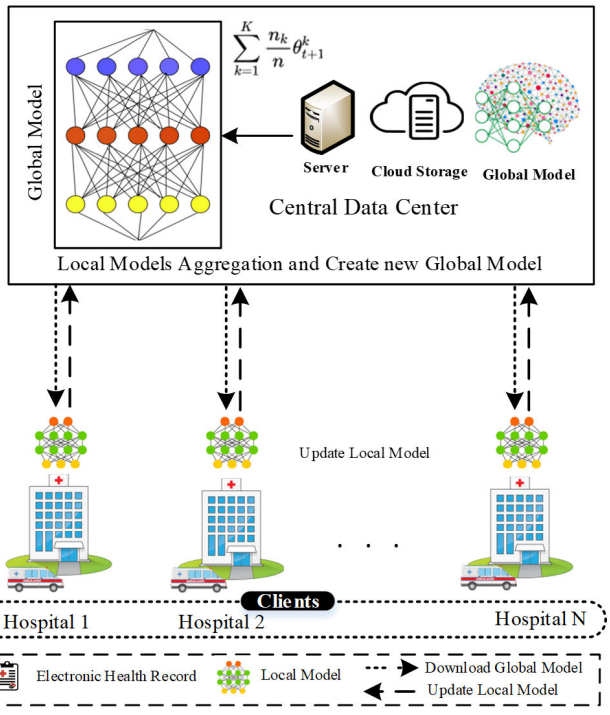


FIGURE 2. Architecture of federated learning in decentralized healthcare.

a few epochs of stochastic gradient descent (SGD) on local data:

$$\theta_k^{(t+1)} = \theta^{(t)} - \eta \, \nabla \ell_k(\theta^{(t)}; B_k), \tag{2}$$

where $B_k \subseteq D_k$ is a local mini-batch and η is the learning rate. It results in an updated local model $\theta_k^{(t+1)}$, which is uploaded to the server for aggregation. *Note:* Some FL variants, such as FedProx [34], modify this update rule by adding a proximal term $\mu \|\theta - \theta^{(t)}\|^2$ to address heterogeneity in the data. The term is absent in the conventional FL formulation.

2) BLOCKCHAIN LOGGING AND VALIDATION

Following local training, clients upload their updates on $(\theta_k^{(t+1)}, n_k)$ to a permissioned blockchain network. A Practical Byzantine Fault Tolerance (PBFT) consensus mechanism [35] is used to verify the authenticity of updates and append them to the distributed ledger. The model parameters can be stored off-chain (for example, using IPFS) with the hashes recorded on-chain to ensure their integrity.

3) GLOBAL AGGREGATION

After collecting enough updates, the aggregator calculates a weighted average:

$$\theta^{(t+1)} = \frac{1}{\sum_{k=1}^{K} n_k} \sum_{k=1}^{K} n_k \, \theta_k^{(t+1)}, \tag{3}$$

known as *Federated Averaging (FedAvg)* [36]. It makes institutions with larger datasets more impactful on the global model, thereby enhancing generalizability and equity.

4) DISSEMINATION

The new global model $\theta^{(t+1)}$ is published on the blockchain. Clients obtain the model and initiate the subsequent iteration, continuing the cycle until convergence or reaching the communication budget.

5) BLOCKCHAIN INTEGRATION

Blockchain constitutes the foundation of this architecture because it (i) provides auditability and immutability of updates to training, (ii) supports decentralized aggregation with BFT consensus, and (iii) provides transparent incentive mechanisms to incentivize participation by size of contribution or quality of model.

The architecture of federated learning in decentralized healthcare comprises multiple participating institutions (e.g., hospitals) connected to a central server, as illustrated in Figure 2. This server orchestrates the communication and aggregation of local models trained on institution-specific data silos, which typically contain EHR and other patient-related information [37]. The federated learning process operates in iterative rounds: the central server broadcasts a global model to all institutions, each institution trains its local model using the global model as a starting point, and then sends the updated parameters back to the server [38], [39]. The server aggregates these updates to create a new global model, which is then disseminated to initiate the next training round. This decentralized approach offers several advantages for healthcare applications, including enhanced patient privacy through on-site data retention, distributed training workload across the network, and fostered inter-institutional collaboration without direct data sharing [40].

C. TYPES OF FL FOR HEALTHCARE

In healthcare, different types of FL addresses unique data distribution challenges across medical institutions and devices. These variations in FL models ensure that sensitive healthcare data can be utilized for advanced model training while maintaining privacy and compliance with regulatory standards. The primary types of FL used in healthcare include Horizontal Federated Learning (HFL), Vertical Federated Learning (VFL), and Federated Transfer Learning (FTL), as shown in Figure 3, each of which caters to specific scenarios, depending on how the data are distributed among the participants, making them vital in different healthcare settings.

In HFL for healthcare, multiple clients collaborate to train a shared global model while maintaining their data locally. This approach is ideal when datasets share the same features but differ in the samples. The key aspects of HFL in healthcare include ensuring a shared feature space across client datasets, each containing unique patient data in its sample space. Clients independently trained the same type of AI model, such as a neural network, on their respective datasets [41].

A central server then aggregates these local updates to generate a global model without requiring direct access to raw data, thereby preserving patient privacy [42]. An illustrative application of HFL in healthcare is speech disorder detection, where users record a specific sentence on their smartphones, representing the shared feature space. At the same time, their diverse voices constitute a varied sample space. Each device trains a local model, and the central server combines these updates to refine the global speech recognition model. This collaborative approach fosters model development while upholding stringent privacy and security standards across healthcare institutions [42].

VFL in healthcare involves collaborative training on datasets that share sample spaces but feature distinct data attributes. Key aspects of VFL include participants working with a shared sample space, such as data from the same patient or case, while contributing different feature spaces that encompass unique attributes or information. Techniques for entity alignment were employed to manage overlapping data samples across participants, and encryption methods were integrated into local training processes to bolster data privacy. This decentralized collaboration framework is particularly well-suited to sensitive domains, such as oncology, cardiology, or mental health, where data is often fragmented across multiple providers. The partnership between a hospital and an insurance company exemplifies a practical application of VFL in healthcare. Here, shared patients constitute the common sample space, with the hospital providing medical records (one feature space) and the insurance company contributing healthcare cost data (another feature space). Collaboratively, they trained an AI model designed to enhance healthcare decision-making by leveraging complementary data while upholding stringent privacy and security measures [43]. This approach fosters the potential for more comprehensive and accurate healthcare insights derived from diverse yet compatible datasets. Mathematically, VFL can be represented as:

$$L_{\text{VFL}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} w_i L(\theta; D_i) \tag{4}$$

where: θ indicates the shared model parameters, D_i represents the local dataset held by participant i, $L(\theta; D_i)$ reflects the local loss computed on D_i, w_i is a weighting factor reflecting the relative weight of each participant's data. As a result of this method, heterogeneous data distributions and multiple domains in the data are leveraged between collaborating entities to facilitate knowledge transfer across domains and enhance learning outcomes [44].

In contrast, FTL is designed for settings where neither the sample nor feature spaces has any overlap. This is common when healthcare institutions from different regions, specialties, or populations aim to train a model collaboratively despite substantial heterogeneity in their data. For instance, in healthcare, multiple hospitals from different areas may collaborate to use FTL to create a predictive model for disease outbreaks. Each hospital contributes unique patient demographics and data structures, necessitating the

use of transfer learning techniques to share knowledge effectively. The resulting model benefits from the diversity of data sources, resulting in enhanced predictive accuracy across different population groups [45]. FTL is particularly suited for modeling rare diseases, where limited samples at each site necessitate the generalization of cross-domain knowledge. Unlike VFL or HFL, FTL requires sophisticated domain adaptation techniques to bridge semantic gaps across datasets.

This approach empowers healthcare institutions to harness collective data insights while safeguarding data privacy and addressing the complexities of heterogeneous datasets. Mathematically, FTL can be represented as:

$$L_{\text{FTL}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(L(\theta; D_i) + \lambda \sum_{j\neq i} T(\theta; D_i, D_j)\right) \quad (5)$$

where: $L(\theta; D_i)$ represents the standard local loss applied to dataset D_i, $T(\theta; D_i, D_j)$ represents the transfer loss function, which records the divergence between the knowledge learned from datasets D_i and D_j. The Degree of Regularization can be determined via distance metrics such as Maximum Mean Discrepancy (MMD), Kullback-Leibler divergence, or feature alignment penalties, where λ is the hyperparameter that controls the degree of regularization. Additionally, meta-learning and adversarial alignment techniques may also be applied to enhance FTL robustness in non-IID healthcare data.

D. INTEGRATION OF BLOCKCHAIN AND FL

In the healthcare sector, the integration of blockchain and FL can revolutionize medical research and patient care by enabling secure and privacy-preserving data sharing. Healthcare providers can collaborate in training AI models for disease diagnosis, treatment planning, and personalized medicine without exposing patient data to unauthorized parties [46]. This approach enhances the accuracy and efficiency of medical AI applications while maintaining patient confidentiality and regulatory compliance. Financial institutions can also benefit from integrating blockchain and FL to enhance fraud detection, risk assessment, and customer profiling. By securely sharing model updates across decentralized servers, banks can improve the accuracy of AI models for fraud prevention without compromising customer data privacy [47], [48]. This collaborative approach enables financial organizations to leverage collective intelligence while safeguarding sensitive financial information. The integration of blockchain and FL presents challenges, such as communication overhead, data heterogeneity, and model aggregation complexities. Efficient communication protocols are essential for minimizing the overhead associated with exchanging model updates between devices or servers. Addressing data distribution variations and ensuring model consistency across diverse data sources requires robust algorithms and aggregation techniques. Overcoming these challenges is crucial for realizing the full potential of BCFL for various applications [49].

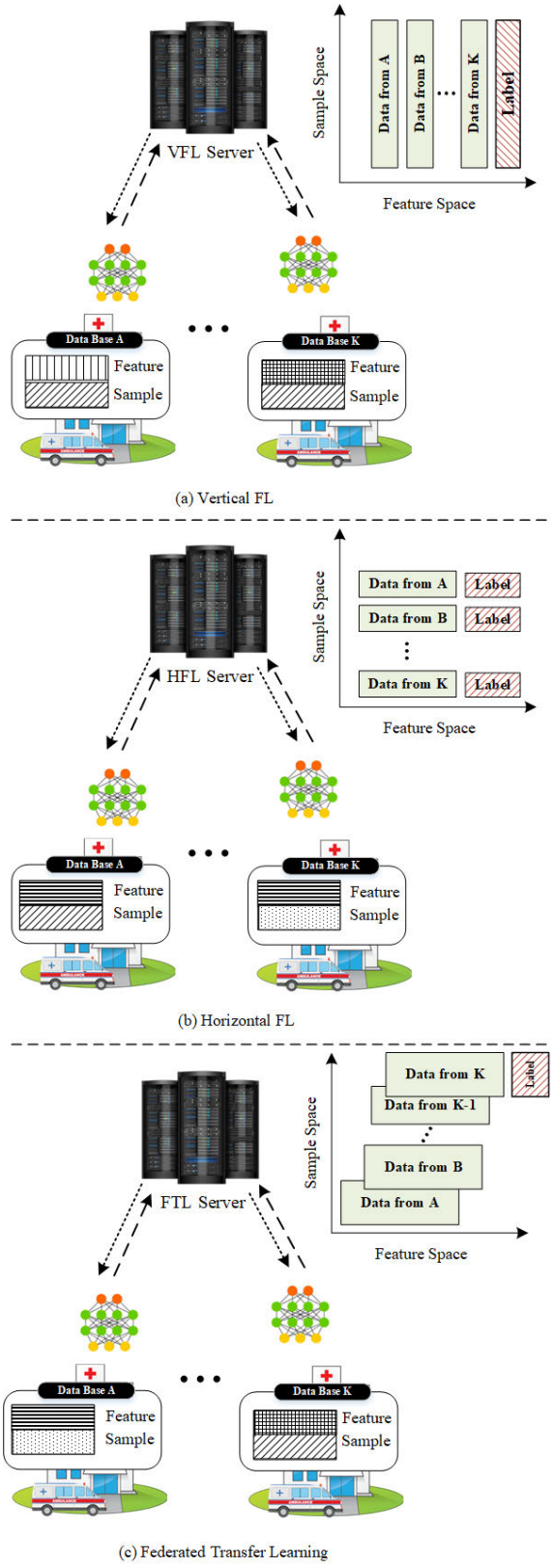


FIGURE 3. Three categories of FL in decentralized healthcare.

The BCFL system integrates blockchain technology with federated learning to create a secure, decentralized, and efficient machine learning framework. This innovative approach involves multiple steps, starting with initialization, where clients set up their environments based on the pre-negotiated parameters stored on the blockchain. Local model training follows with each client using their private dataset to train the assigned model. Upon completion, clients upload their trained local models to the blockchain system, often using reference links because of the large size of these models. The blockchain then broadcasts these transactions for cross-validation across the network [50]. A designated node, selected through a consensus protocol, aggregates the local models to update the global model and generates a new block. This updated global model is then shared with all clients through a block broadcast. Clients download the latest worldwide model and begin a new iteration of training, repeating the process until the predefined end conditions are met. This workflow ensures data privacy, model integrity, and collaborative learning across distributed participants by leveraging the strengths of both federated learning and blockchain technology [51].

The high-level workflow of BCFL is illustrated in Figure 4. This workflow begins with a task publisher initiating a service request and deploying a smart contract to govern a federated learning task. Participants then enroll in the smart contract, download the global model, and train it using their local datasets. The smart contract facilitates interactions among the blockchain network, FL participants, and miners, with participants uploading their local model updates to miners for validation. Miners engage in a consensus algorithm to generate a new block containing the updated global model, which is then disseminated across the network [52]. This process is iterated until the model converges or reaches the designated federation round, ensuring a secure, decentralized, and efficient approach to collaborative machine learning. As shown in Figure 5, multiple hospitals collaborate through the blockchain to run DL algorithms using local EHR data.

III. BLOCKCHAIN ENABLED FL FOR SECURITY AND PRIVACY PROTECTION

Blockchain-FL provides a viable solution to guarantee security and privacy protection in federated machine learning frameworks. By utilizing blockchain's open and distributed property, FL systems can guarantee data integrity, confidentiality, and accountability during model training. Blockchain incorporation into FL diminishes the risk caused by centralized storage and processing of data and therefore the risk of data breach and illicit access to sensitive information. Such integration facilitates secure and privacy-preserving collaboration among machine learning processes among scattered devices or servers, building trust and openness in sharing data [53]. Decentralized BCFL supports evading single points of failure and maliciousness by spreading data processing and model updating among network nodes. Such architecture increases the resilience and security of FL frameworks, making them even more attack-resistant and immune to data tampering. By facilitating an auditable and secure record book for model updates and data exchanges, blockchain technology guarantees the traceability and integrity of the complete FL process, and thus improves accountability and adherence to data privacy law. In healthcare, BCFL can transform data exchange and collaborative research on EHR while protecting patient privacy [54].

Medical professionals can safely develop AI models using secret healthcare information without revealing a patient's personal information, enabling proper diagnostics and tailored treatment advice. Blockchain and FL in the healthcare sector provide guarantees that data privacy and security are assured, hence enabling innovation and breakthroughs in medical research and patient care. BCFL can also be beneficial to banks for privacy and security protection while conducting fraud detection, risk assessment, and customer profiling. Banks are able to collaborate and train AI models to prevent fraud without breaching customer data privacy through the utilization of the decentralized and tamper-resistant nature of blockchains. This method makes financial AI products more accurate and efficient, and confidential sensitive financial information, with a lower risk of fraud and data breaches [55]. Scalability, interoperability, and compliance are challenges of BCFL for security and privacy protection. Scalability problems are created by the great quantity of transactions and updates within blockchain networks and may have an effect on the performance of FL systems. Interoperability problems are created by the requirement to ensure data exchange between various blockchain platforms and FL frameworks is seamless and efficient, and that can be ensured by standardized protocols and data formats. Regulatory compliance is most important in addressing legal and privacy issues while releasing data and training models, especially in strictly regulated sectors like finance and healthcare. FL and blockchain technology provide a strong combination to improve the security, privacy, and coordination of healthcare data [53]. The decentralized and privacy-focused nature of FL renders it most suitable for application within the healthcare setting, where data privacy is crucial. Integration of blockchain technology with FL further increases security and transparency in sharing and training healthcare data and models [55].

The combination of FL and blockchain technology has been the main concern in improving the security and privacy of healthcare data management. Recent research by [56] proposed the BCFL framework to boost the security and privacy of EHR further. The suggested BFL framework is efficient in integrating blockchain technology with FL so that hospitals can enjoy local control of data as well as participate in a cooperative model training process. The suggested framework achieved improved accuracy 92.5% on the Chronic Kidney Disease (CKD) dataset compared to baseline techniques in terms of accuracy and loss measures. Moreover, the research gave high value to off-chain storage systems for data, like the Interplanetary File System (IPFS),

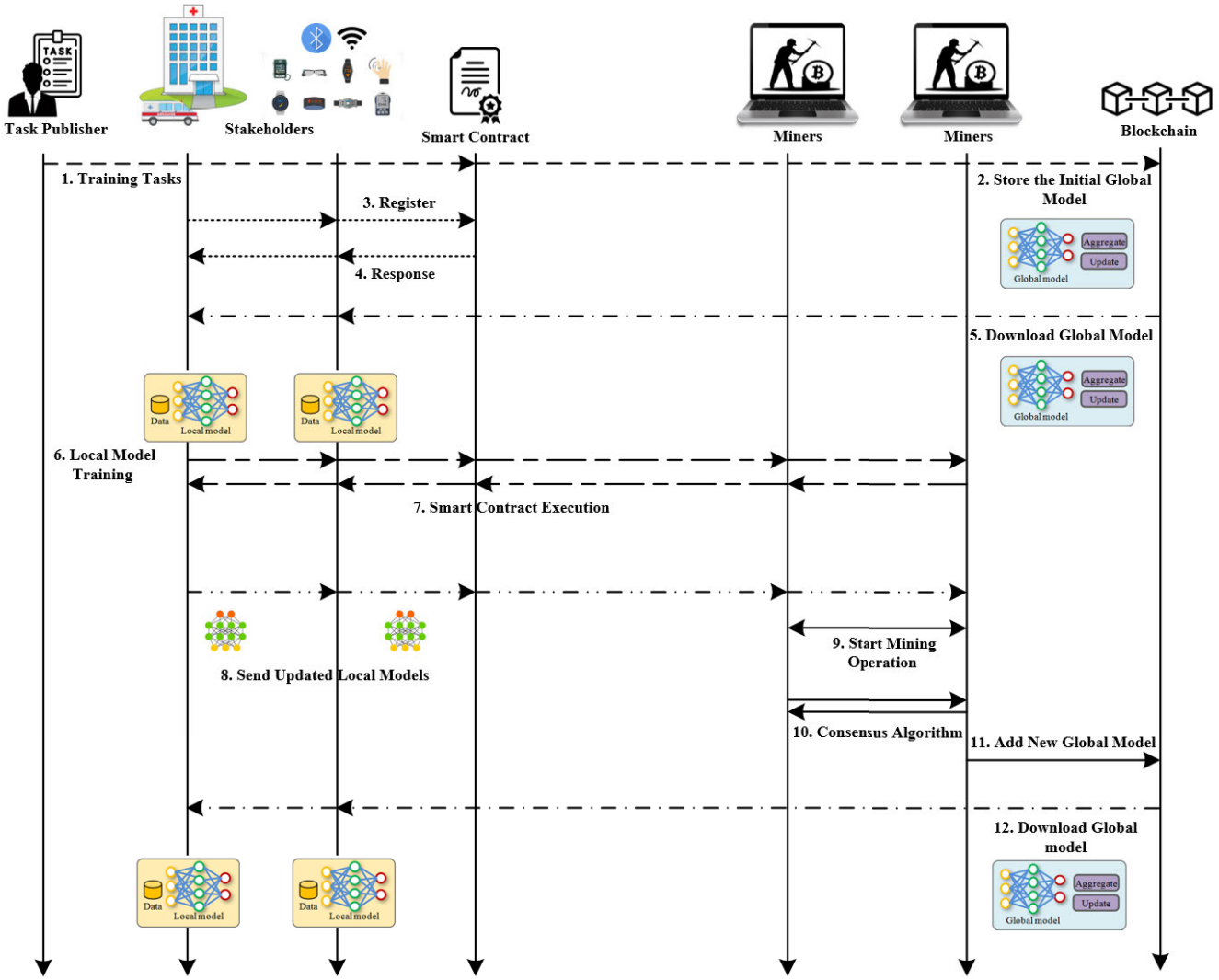


FIGURE 4. High-level workflow of BCFL.

where EHR data is stored securely. Another work by [57] also suggested a comparable framework that unifies FL and blockchain to secure medical records, specifically medical imaging for disease diagnosis, e.g., COVID-19. The framework is a decentralized training model that is exchanged with a federated server and implements token-based authentication to ensure secure data sharing.

In [58], a secure healthcare system combining blockchain, FL, and cloud computing was proposed. This system uses proxy re-encryption and the Elliptic Curve Digital Signature Algorithm (ECDSA) to ensure secure data sharing and prediction. The integration of blockchain facilitates a verification system for FL model aggregation, thereby enhancing data integrity and authenticity. The performance analysis demonstrated the efficiency of FL training and model evaluation, addressing key security concerns in healthcare data management. The work in [59] focused on collaborative federated learning (CFL) for multi-modal COVID-19 diagnosis at the edge. This study highlights the importance of privacy-aware collaborative learning frameworks in healthcare. The proposed system employs federated averaging with a weighting factor to update model weights and integrates hardware optimization techniques for edge deployment. Adversarial robustness and privacy are crucial aspects discussed in this context, highlighting the need for secure and efficient machine learning deployment in healthcare settings. Reference [60] introduced a blockchain-based federated learning framework specifically designed to secure the global model used for disease classification. The framework incorporates SMPC for encrypted inference and model verification, ensuring the privacy and integrity of local models. This decentralized approach mitigates the risks associated with malicious attacks on local models, enhancing the overall security and robustness of the federated learning system. In [61], a blockchain-based personalized FL system was proposed for the IoMT devices, aiming to address the privacy

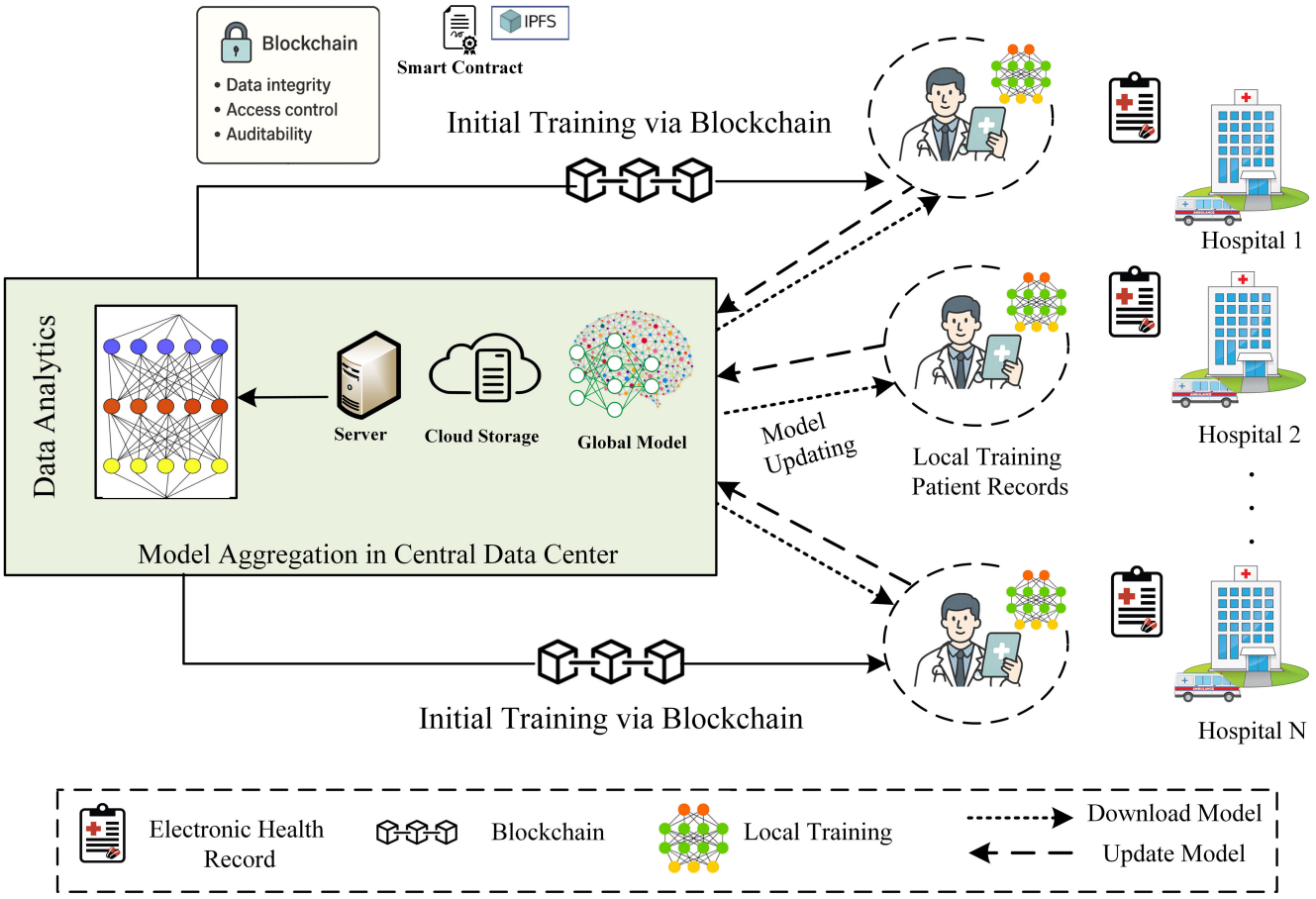


FIGURE 5. Decentralized healthcare data analysis through federated learning and blockchain integration.

and security concerns of medical data. By incorporating base and personalization layers, the system demonstrated superior performance in non-IID scenarios, offering better results on heterogeneous medical datasets compared to traditional approaches. The study emphasized the significance of decentralized architectures and showcased the potential of combining FL with blockchain to enhance privacy protection and mitigate risks of single points of failure. Reference [62] explored the use of FL in a fog-cloud environment for fraud detection in healthcare applications. The proposed FL-BETS system leveraged blockchain technology to enable secure data sharing and privacy preservation. Through dynamic machine learning methods, the study addressed critical issues, including fraud detection, resource leakage, and energy efficiency. The results highlighted the effectiveness of the proposed system in maintaining privacy and preventing fraud, demonstrating its potential to optimize resource utilization in IoMT networks. The work in [63] introduced a security model integrating Convolutional Neural Networks (CNN) with BCFL to protect patient data privacy and detect malicious users. The proposed CNN_BC_Cryp_FL model achieved significant improvements in accuracy, precision, recall, scalability, and data privacy. By focusing on the preservation of electronic health record privacy preservation and the identification of abnormal users, the model effectively safeguarded sensitive patient information, showcasing the potential of combining CNNs with blockchain for enhanced security in healthcare systems. In [64], a novel framework combining BCFL and capsule networks was proposed for the detection of COVID-19 patients. The framework utilizes data normalization techniques to handle heterogeneous data from various hospitals, ensuring privacy and accuracy in diagnosis. This study introduced a new dataset, CC-19, and demonstrated the effectiveness of the framework in processing big data, securing data sharing, and training collaborative models. The results indicate high accuracy and efficiency in detecting COVID-19 patterns, highlighting the benefits of integrating blockchain with FL for secure and accurate medical diagnoses. Reference [65] focused on enhancing the security and privacy of EHR stored in the cloud using a BCFL approach. By combining lightweight encryption with FL, the proposed model ensures secure access to patient data while preventing unauthorized access. This study evaluated the encryption and decryption times for various key sizes and compared the proposed technique with existing algorithms. The results demonstrated an improved security performance,

emphasizing the importance of innovative security measures in cloud-based healthcare environments. Several studies have investigated the combination of FL and blockchain to enhance the trustworthiness, security, and efficiency of healthcare systems.

References [66], [67], [68], and [69] discuss the integration of blockchain in federated learning frameworks to address centralization and security issues, emphasizing the potential for these technologies to transform evidence-based medicine and improve patient outcomes through enhanced data provenance and incentivization mechanisms. This study highlights the application of machine learning algorithms, including deep learning, support vector machines, and decision trees, in healthcare domains such as genomics, mental health, and medical imaging, while also noting the challenges related to data privacy and model biases. Similarly, [70] introduces the HealthFed framework, which combines Secure SMPC with FL and blockchain for privacy-preserving collaborative learning among healthcare providers. The framework was shown to outperform centralized approaches in breast cancer detection, achieving high precision, accuracy, and F1 scores while ensuring the privacy of sensitive medical data. The use of SMPC for local model aggregation and blockchain for secure data sharing underscores the potential of these technologies to revolutionize healthcare applications.

Reference [71] extends the application of blockchain and FL to wearable IoT devices for predictive healthcare services. By leveraging machine learning algorithms and private blockchain technology, this study aimed to secure healthcare IoT networks and enhance the accuracy of real-time patient monitoring. The results demonstrate the effectiveness of this approach in achieving high testing accuracy and maintaining the integrity of the healthcare data. Reference [72] proposes the FLED-Block (Federated Learning Ensembled Deep Learning Blockchain) framework, which integrates FL, deep learning models, and blockchain for COVID-19 diagnosis using CT image datasets. This study demonstrates the superior performance of the proposed model in terms of accuracy, precision, recall, and F1-score and emphasizes the importance of secure data retrieval and sharing in federated learning environments. The need for improvements in handling real-time clinical databases was also highlighted, indicating areas for future research. The emergence of FL in healthcare informatics has been thoroughly discussed in a study that highlights the benefits of decentralized learning models over traditional centralized models. This study examines the applications of FL in IoT-assisted healthcare, with a focus on data privacy, security, and integration with advanced communication networks [73]. Key frameworks, such as TensorFlow Federated (TFF) and LEAF, are examined, emphasizing the importance of decentralized learning to ensure data integrity and privacy. The study also proposed a blockchain-envisioned signcryption scheme for secure data transmission in healthcare IoT ecosystems, achieving a computing accuracy of 95% through multiple rounds of federated averaging using CNN, Long Short-Term Memory (LSTM), and multi-layer perceptron (MLP) models [74]. Another study [75] introduces the FIDChain (Federated Intrusion Detection System for Blockchain) algorithm, which integrates federated learning and blockchain technology to enhance intrusion detection in IoT healthcare systems. The proposed system demonstrated a high accuracy 99.9% and a low false alarm rate, demonstrating its effectiveness in detecting botnets in IoT applications. By combining machine learning models with blockchain, the study addresses security challenges, ensuring transparency and protection against poisoning attacks in healthcare data. The architecture of the FIDChain system significantly improves the detection response time and data privacy, highlighting its potential for securing healthcare systems [75].

To address data sparsity and efficiency issues in healthcare applications, Cross-Cluster Federated Learning (CFL) has been proposed to connect Blockchain-Based Federated Learning clusters, thereby improving data exchange and system efficiency. The CFL framework leverages blockchain technology to secure data exchange and model aggregation across clusters, thereby enhancing the privacy and effectiveness of machine learning models in IoMT. The proposed system demonstrated significant improvements in model accuracy and convergence rates, paving the way for robust and privacy-preserving healthcare applications. The proposed framework emphasizes the secure sharing of patient health records through blockchain, addressing challenges such as the processing requirements for small devices. The study highlights the potential of combining these technologies to improve healthcare delivery and predictive modeling [76]. A comprehensive solution for COVID-19 monitoring was developed using deep learning algorithms integrated with blockchain technology. The proposed system architecture includes QR code-based health profiles, blockchain transactions, and off-chain data storage to ensure data provenance, privacy, and security. The study addresses challenges in federated learning, such as non-iid data and communication delays, and introduces novel contributions, including blockchain-based trust and Intel SGX TEE security. The system demonstrated promising outcomes in COVID-19 monitoring and highlighted the potential for broader adoption of IoHT-based health systems [77]. Table 3 provides a comprehensive summary of recent studies on blockchain-integrated federated learning, focusing on methods, datasets, the role of blockchain, and the strengths and limitations.

A. QUANTITATIVE EVALUATION OF BCFL FRAMEWORKS

BCFL architectures provide new robustness and decentralization properties for model training. They come at the cost of performance, scalability, and implementation complexity, though. The following section provides a systematic review of BCFL systems across four dimensions: model performance, robustness and privacy, efficiency and scalability, and methodological consistency.

TABLE 3. Taxonomy of blockchain-based federated-learning studies in healthcare.

FL Method	Blockchain Role	Dataset(s)	Strengths	Limitations	Comp. OH	Comm. Cost	Scalability limits	Ref
HFL	EHR privacy/security	CKD	92.5% acc. on EHR data	Dataset size	Low	Low–M	4-device demo	[46]
HFL	Smart-contract access control	Medical images	Robust COVID-19 detection	Latency challenges	Moderate	High	3 fog nodes	[47]
HFL	Immutable ledger verification	Medical records	Secure ECDSA aggregation	Perf. overhead	High	Moderate	5 devices	[48]
CFL	Secure data sharing	Imaging + records	↑diagnostic accuracy	Efficiency issues	High	High	$\leq$100 tx/s	[49]
VFL	Global model + local chains	OCTMNIST, PneumaMNIST	+25 / +10% acc.	Malicious clients	Moderate	Moderate	50 clients	[50]
VFL+PFL	Privacy + personalisation	Fashion-MNIST, CIFAR-10	PFL boosts acc.	Scalability issues	Moderate	High	50-client limit	[51]
FTL	Fraud / anomaly on edge	Synthetic healthcare	↑accuracy at edge	Efficiency issues	Moderate–High	Moderate	20 IoT nodes	[52]
VFL	Verifiable FL	DB1, DB2	82.5 / 75.5 % acc.	Costs NR	Low–Moderate	Low–Moderate	NR	[53]
VFL	Privacy-preserving CT	CC-19	80 sens., 90 spec.	Data-leak risk	Moderate	High	30 clients	[54]
VFL	Decentralised learning	PH records	↑security vs baseline	Efficiency issues	Moderate	Moderate	NR	[55]
VFL	Trust / governance	Mixed medical + genomic	Efficient aggregation	Privacy/fairness	High	Moderate	TPS bottleneck	[56]
VFL	SMPC-assisted	Breast cancer	98 F1 in 10 iters	Enc. overhead	High	Very High	Small set	[57]
VFL	IoT healthcare	Skin / monitor	97.7 % acc.	IoT compat. issues	Low–Moderate	Low	Device limits	[58]
VFL	Secure HAR	HAR	90.4 % acc.	Needs a robust net.	Moderate	Moderate–High	Bandwidth bound	[59]
VFL	Secure COVID-CT	CT image	97.3 % acc.	Training time	High	Moderate	5 hospitals	[70]
VFL	Vehicular SpO_2	Pulmonary records	97 % acc.	Data consistency	Moderate	High	Density-dependent	[71]
HFL	Transparent EMR sharing	EMR-CKD, MIMIC-III	99.99 % acc.	Security hardening	Moderate	Moderate	NR	[72]
VFL	Data integrity	MIMIC local	97.6 % acc.	Centralised bottleneck	High	Low	Single aggregator	[73]
CFL	Split-learning IoMT	Split MNIST, IFO	↑privacy, ↑acc.	↑comp. complexity	High	High	Cluster-size bound	[74]
HFL	IoT diabetes	Pima1/2	81–83 % acc.	Small-device limits	High	High	Edge CPU/bw	[75]
VFL	Parkinson's 5.0	PPMI, PAM50	>95 % acc.	Comp. complexity	High	Moderate–High	Chain+FL scale	[76]
HFL	QoS-aware IoHT	Clinical, ICU	3.95 / 36.5 % OH savings	Comm. delays	**Low**	Reduced	10 nodes; NR	[77]

Abbreviations: L = Low, M = Moderate, H = High, NR = Not reported, OH = Overhead; "acc." = accuracy; "tx/s" = blockchain transactions per second.

1) MODEL PERFORMANCE AND CONVERGENCE BEHAVIOR

Model performance is the main indicator of BCFL performance. Models like BPFL have 92% to 98% classification accuracy on datasets like MNIST and CIFAR-10 [79], [80]. The use of blockchain consensus protocols can, however, add 10–15% to convergence time because of block validation delay mostly. In Byzantine environments, Byzantine-resilient protocols such as BRFL are effective with 85.3% accuracy on CIFAR-10 in 30% malicious clients, compared to a high of 18% [81] in non-blockchain FL. Hierarchical configurations, such as T-FedHA, are also advantaged by reduced communication overhead. These schemes reduce communication rounds by as much as 40% and have similar accuracy (89.7%) on high-dimensional data, i.e., medical imaging [82]. In contrast, fully decentralized FL systems suffer 15–20% performance loss due to the absence of updated model updates and staleness in asynchronous structures [83], [84].

2) ROBUSTNESS, SECURITY, AND PRIVACY TRADE-OFFS

Blockchain-enabled federated learning improves adversarial robustness via reputation-based verifiability and tamper-proof logging. For example, BRFL uses blockchain-referenced reputation scoring to identify over 95% of gradient-poisoning attacks with accuracy decline limited to under 5% even under high malicious participation [85]. Defendroid achieves 94.5% accuracy in Android app adversarial detection via explainable AI methods, lowering false positives by 22% compared to non-blockchain alternatives [86].

Privacy-enhancing techniques particularly differential privacy (DP) provide a trade-off between utility and privacy. DP methods in BCFL minimize data loss by 50% but 8–12% [84], [87] lower model accuracy. TEEs-based multi-component designs like VeryFL optimize training efficiency by 18% with 99% data integrity by secure enclave execution [81].

3) EFFICIENCY AND SCALABILITY CONSIDERATIONS

Consensus algorithms play a significant role in BC-FL energy and time efficiency. Proof-of-Work consensus adds an additional 30–45% of energy use over less burdensome protocols such as Practical Byzantine Fault Tolerance (PBFT) [88]. Sharded blockchain structures enhance scalability through decreasing aggregation time by 55% in big systems, but cause about 6-9% loss of accuracy through partial gradient sharing [89].

A comparison of BCFL frameworks in real-world scenarios is presented in Table 4 with a focus on healthcare use cases. FLED-Block and PQS-BFL are notable as they surpass the 95% accuracy rate necessary for clinical adoption in decentralized environments. DP-BIoT-FL attains sub-minute latency in private Ethereum deployments, while PQS-BFL provides cryptographic scalability with

TABLE 4. Performance summary of blockchain-FL frameworks for healthcare applications. Abbreviations: HFL = Horizontal Federated Learning; PQC = Post-Quantum Cryptography; n/r = not reported.

Study / Year	FL Type	Dataset & Task	Top Metric	Latency (s)	Gas / Tx	Blockchain Mechanism
FLED-Block (2022) [70]	HFL	Multi-hospital CT (COVID-19)	98.2% Accuracy	n/r	n/r	PoW ledger records model hashes; ensemble of ELM and CapsNet
DP-BIoT-FL (2024) [90]	HFL	SVHN (IoT health analytics)	64.5% Accuracy	6.0 ± 0.3	22,152 gas	Ethereum + IPFS; smart contract audit trail with differential privacy
PQS-BFL (2025) [91]	HFL + PQC	MNIST (robustness study)	98.8% Accuracy	4.8 ± 0.4	1.72×10^6 gas	Dilithium (ML-DSA-65) signatures; on-chain PQC verification
FBS-Health (2023) [92]	Federated Ledger	EHR operations (benchmark)	n/r	0.068–19.0	n/r	Hyperledger (permissioned); query latency 68–100 ms, writes < 20 s

post-quantum signatures at the cost of increased gas expense. Both of these incur overheads of less than 0.02% of overall training time, so they are practically feasible.

B. SECURITY THREATS AND DEFENCE MECHANISMS

FL allows for privacy-preserving model training; however, its decentralized update pipeline enlarges the adversarial surface. When FL is combined with blockchain, the immutability of the ledger and the economic incentives associated with it reduce, but do not eliminate, security risks. The purpose of this section is to provide an overview of the significant threats and to synthesize contemporary countermeasures with a focus on healthcare deployments, where confidentiality and integrity are of utmost importance.

1) ATTACK LANDSCAPE

- **Model- and data-poisoning.** Clients with malicious intentions can introduce backdoors or craft gradients that degrade overall accuracy or result in targeted misclassification. Temporal analysis of client trajectories, as implemented in *FLDetector*, exposes such behavior with high precision [78].
- **Inference attacks.** Membership and property inference techniques seek to identify the presence of specific records or sensitive attributes in the training set. A recent survey indicates that there are more than forty gradient-based extraction methods [93].
- **Sybil and free-rider attacks.** The creation of pseudo-identities benefits adversaries by giving them disproportionate voting power, corrupting consensus, and reaping rewards without actually contributing to the standard [94].
- **Denial-of-Service and staleness flooding.** It is possible for attackers to overwhelm aggregators or miners with outdated or malformed updates, slowing convergence or even halting training altogether.
- **Blockchain-layer threats.** It is possible for blockchain-specific issues, such as 51% attacks or smart contract re-entrance, to compromise reward logic and auditability.

2) PRIVACY- AND INTEGRITY-ENHANCING MECHANISMS

a: ROBUST AGGREGATION AND ANOMALY DETECTION

Forecasting of benign gradient paths over time in combination with trimmed-mean or coordinate-median rules flags poisoned updates before on-chain commitment, cutting attack success rates by up to 92% [78].

b: DIFFERENTIAL PRIVACY (DP)

Local Gaussian noise calibration mitigates inference risk without compromising clinical utility. A new breast-cancer example had 96% accuracy at privacy budget ε=1.9 [97].

c: END-TO-END HOMOMORPHIC ENCRYPTION (HE)

Ciphertext-only aggregation on-chain, as shown in a Pattern-based prototype, incurs less than 2% performance loss but eliminates server-side plaintext exposure [98].

d: SECURE MULTI-PARTY COMPUTATION (MPC)

Secret-sharing schemes perform aggregation operations without ever reconstructing a full key, eliminating a prime compromise target [99].

e: FEDERATED DISTILLATION

Sharing class logs instead of full weights reduces the attack surface and conserves bandwidth. Selective-FD further cleans noisy knowledge and outperforms vanilla FL under highly non-IID data [100].

f: BLOCKCHAIN-NATIVE INCENTIVES

Peer-to-peer voting and reward-slash smart contracts redirect staked tokens away from adverse behavior or Sybil nodes, matching economic incentives to honest behavior [94]. Together, these defenses indicate that high privacy guarantees and Byzantine resistance are achievable at non-prohibitive accuracy or latency costs, paving the way for safe, scalable BCFL deployments in healthcare.

C. FAIRNESS IN FEDERATED HEALTHCARE SYSTEMS

Non-identically distributed data across healthcare institutions poses significant challenges to the accuracy and fairness of FL models. Heterogeneities in data, such as class imbalance, demographic disparities, and non-uniform feature distributions, have the potential to cause biased local updates and degrade global performance. Experiments show accuracy drops of up to 15% in non-IID environments. To mitigate such challenges, new solutions like EquiFL [95] and FairGrad [96] present fairness-aware optimization, whereas techniques like clustered FL and knowledge distillation enhance model personalization and help reduce inter-site inconsistencies.

Although progress has been made, significant challenges remain. Optimizations to enhance fairness tend to be less accurate, and privacy solutions such as differential privacy inadvertently increase bias. Personalized DP, secure aggregation, and adversarial debiasing have high potential but require clinical trials to validate their effectiveness. Additionally,

there is no unified standard measure for fairness, making it more challenging to evaluate. Scalable, ethically sound FL protocols that adjust for institutional heterogeneity while providing privacy and fairness guarantees are needed in upcoming studies [95], [96].

IV. APPLICATIONS

This section discusses the various applications of BCFL in healthcare. These applications include Medical Diagnostics, Precision Medicine, Disease Outbreak Prediction, Drug Discovery and Development, Healthcare Resource Allocation, and Personalized Treatment Plans. The effectiveness of BC-enabled FL in healthcare applications can be evaluated based on Data Privacy, Collaboration Scale, Real-time Capability, Patient Impact, and Implementation Complexity. Table 5 lists different applications and their corresponding evaluation criteria.

A. MEDICAL DIAGNOSTICS

Medical diagnosis involves the identification of diseases based on the patient's symptoms. Traditionally, this process takes place at health centers or hospitals, where doctors physically examine patients. However, in emergencies, this method can delay treatment and potentially lead to loss of life. Telemedicine was introduced to address this issue, allowing remote patient monitoring. Additionally, advancements in AI have significantly impacted healthcare, with machine learning models enhancing disease diagnosis by utilizing training data from various healthcare centers [33]. Despite its benefits, telemedicine has a notable drawback: patients may struggle to accurately convey their symptoms or the severity of their condition without in-person interactions with healthcare professionals.

In [101], researchers developed a framework that utilized BCFL to identify 15 distinct lung diseases with high accuracy. This system enables multiple healthcare providers to contribute to a shared diagnostic model without exchanging sensitive patient data directly. The blockchain component ensures a tamper-proof and transparent record of all model updates and transactions, fostering trust among participating institutions. The application of BCFL in medical diagnostics extends beyond lung disease. Studies have shown its effectiveness in various areas, including breast cancer detection, cardiovascular disease prediction, and neurological disorder diagnosis [101]. These systems can help develop more robust and generalizable diagnostic models by leveraging diverse datasets from multiple institutions. Furthermore, the decentralized nature of BCFL addresses concerns related to data ownership and control, allowing healthcare providers to maintain custody of their patient data while still contributing to collaborative research efforts. This approach not only enhances the accuracy and reliability of medical diagnostics but also promotes a more equitable and inclusive healthcare ecosystem by enabling smaller institutions to participate in cutting-edge research and model development.

B. PRECISION MEDICINE

Blockchain-enabled FL has emerged as a powerful tool for advancing precision medicine by unlocking valuable insights from EMR while maintaining patient privacy and data security. This innovative approach enables healthcare providers to collaborate in developing personalized treatment plans while maintaining the confidentiality of sensitive patient information. By leveraging blockchain technology, the system ensures that patient data remains confidential, tamper-proof, and traceable, thereby addressing critical concerns in healthcare data sharing. Simultaneously, FL allows for the collaborative training of sophisticated machine learning models that can predict patient-specific treatment outcomes with high accuracy. This combination of technologies has shown promise in various areas of precision medicine, including cancer treatment optimization, rare disease diagnosis, and pharmacogenomics [102].

The application of BCFL in precision medicine extends beyond the individual treatment plans. It has the potential to revolutionize clinical trial design and execution by enabling the identification of suitable patient cohorts across multiple institutions, without centralizing sensitive data. Furthermore, this approach facilitates the integration of diverse data types, including genomic, proteomic, and metabolomic data, along with traditional clinical data, leading to more comprehensive and nuanced patient profiles [103]. By allowing healthcare institutions to contribute to shared models while retaining control over their data, BCFL promotes a more equitable and collaborative research environment. Democratization of data access and analysis capabilities has the potential to accelerate the pace of discovery in precision medicine, ultimately leading to more effective and personalized treatments for a wide range of medical conditions [102], [103].

C. DISEASE OUTBREAK PREDICTION

FTL, a variant of Federated Learning, is particularly effective in predicting disease outbreaks across diverse populations. This approach allows multiple hospitals in different regions to collaborate in building robust predictive models for disease outbreaks, such as COVID-19. Each hospital contributes unique patient demographics and medical records, and transfer learning techniques are employed to enhance the model's performance across these diverse datasets. By leveraging blockchain technology, the data-sharing process is secure and transparent, thereby fostering trust among participating institutions. Blockchain ensures that all data transactions are recorded in a decentralized and immutable ledger, which is crucial for maintaining data integrity and preventing tampering [104]. During the COVID-19 pandemic, the application of blockchain-enabled FTL proved invaluable. Hospitals and research institutions worldwide can share insights and data without compromising patient privacy. For instance, blockchain-enabled FTL has facilitated the development of predictive models that could forecast COVID-19 surges, enabling healthcare systems to prepare and allocate resources

more effectively [105]. This collaborative approach also enabled the rapid identification of potential hotspots and the implementation of targeted interventions. Blockchain integration ensured that all contributions were transparent and verifiable, which was essential for maintaining public trust and ensuring the accuracy of predictive models. Studies have shown that such a decentralized and secure framework can significantly enhance the responsiveness and effectiveness of public health strategies during pandemics [106].

D. TELEMEDICINE

In telemedicine, BCFL provides a secure and privacy-preserving framework for remote patient monitoring and diagnosis. This approach allows healthcare providers to collaboratively train models on patient data collected through telemedicine platforms without compromising data privacy. Blockchain ensures that all data transactions are recorded in a decentralized and immutable ledger, thereby enhancing the security and reliability of telemedicine services [107].

E. DRUG DISCOVERY AND DEVELOPMENT

Blockchain-enabled FL has emerged as a transformative approach in drug discovery and development, addressing key challenges in the pharmaceutical industry. This innovative methodology enables pharmaceutical companies and research institutions to collaborate in training sophisticated AI models for drug discovery without directly sharing sensitive data. The integration of blockchain technology ensures a secure, transparent, and immutable record of all contributions, fostering trust among participants while preserving intellectual property rights. This approach has demonstrated significant potential for accelerating the identification of promising drug candidates and reducing the overall costs of drug development pipelines [108].

One of the primary advantages of BCFL in drug discovery is its ability to leverage diverse datasets from multiple organizations while maintaining data privacy and security. For instance, [109] demonstrated how this approach could be used to develop more accurate predictive models for drug-target interactions by combining data from multiple pharmaceutical companies without compromising proprietary information. The decentralized nature of the blockchain ensures that each participant retains control over their data while contributing to the collective knowledge base. This has led to more robust and generalizable models capable of identifying novel drug candidates that might have been overlooked in siloed research environments [109], [110].

F. HEALTHCARE RESOURCE ALLOCATION

BCFL offers a transformative approach to optimizing healthcare resource allocation by predicting patient influx, forecasting equipment and supply needs, and optimizing staff scheduling based on the expected demand. This innovative system allows hospitals to collaborate in training predictive models without compromising patient privacy or competitive advantages. By leveraging blockchain technology, the system ensures that all data transactions are secure, transparent, and immutable, fostering trust among participating institutions and enhancing the overall efficiency of resource management.

A key benefit of BCFL in healthcare resource allocation is its ability to predict patient influx across multiple hospitals. By analyzing historical patient data and current trends, the system can forecast the number of patients expected to visit each hospital. This predictive capability allows hospitals to prepare adequately, ensuring that they have sufficient staff, equipment, and supplies to meet anticipated demand [111]. For instance, during the COVID-19 pandemic, such predictive models could have been invaluable in managing the surge in patient numbers, helping hospitals allocate resources more effectively and avoid overwhelming their facilities. In addition to predicting patient influx, BCFL can forecast the equipment and supply requirements with high accuracy. By integrating data from various sources, including patient records, inventory levels, and usage patterns, the system can predict the future requirements for medical equipment and supplies [112], [113].

G. PERSONALIZED TREATMENT PLANS

Blockchain-enabled FL has revolutionized the development of personalized treatment plans by enabling healthcare providers to create more accurate and individualized care strategies. This approach leverages diverse patient data from multiple institutions, thereby allowing the models to learn from a wide range of patient populations. By integrating blockchain technology, the system ensures that individual patient data remains protected at its source, thereby addressing critical privacy concerns [114]. The decentralized nature of blockchain also ensures the integrity and provenance of the data used in decision-making, which is crucial for maintaining trust and transparency in the healthcare ecosystem. A significant advantage of blockchain-enabled FL in personalized treatment plans is its ability to predict treatment efficacy based on similar cases from multiple sources. In [115], the authors demonstrated how this approach could be used to develop personalized cancer treatment plans by analyzing data from various oncology centers. The federated learning model could identify patterns and treatment responses across different patient demographics, leading to more tailored and effective treatment strategies. Blockchain technology plays a crucial role in ensuring that all data transactions are secure and transparent, fostering collaboration among institutions while maintaining patient confidentiality [116].

Moreover, BCFL addresses the issue of bias in predictive models, which is particularly important in personalized medicine. Bias in healthcare data can lead to unfair and inaccurate predictions that disproportionately affect certain patient groups. By integrating blockchain, the system can implement a bias-mitigation process, ensuring that the

TABLE 5. Technical aspects of BC enabled FL for healthcare applications.

Application	Data Privacy	Collaboration Scale	Real-time Capability	Patient Impact	Implementation Complexity
Medical Diagnostics	H	M	M	H	M
Precision Medicine	VH	M	M	VH	H
Disease Outbreak Prediction	H	VH	H	H	VH
Drug Discovery and Development	VH	H	L	M	H
Healthcare Resource Allocation	H	H	H	M	M
Personalized Treatment Plans	VH	M	M	VH	H
L Low Significance	M Medium Significance		H High Significance	VH Very High Significance	

TABLE 6. Production-level BC-FL deployments in healthcare.

Project	Sites	Domain	Key Gain	BC/FL Stack
MELLODDY [119]	10 pharma	QSAR discovery	+26 % hit-rate	Quorum + FedAvg
FedRMD-CXR [120]	40 (sim.)	Diagnostics	AUROC ↑0.08	Fabric + FedRMD
TNBC-FL [121]	6 hospitals	Oncology	C-index ↑0.05	Fabric + FedAvg
Retina-BCFL [122]	8 clinics	Ophthalmology	AUC 0.970	Fabric + custom

models are fair and equitable. Reference [117] highlighted how BCFL can improve the fairness of predictive models in healthcare by providing an accountable and transparent bias-mitigation process. This approach not only enhances the accuracy of personalized treatment plans but also promotes health equity by ensuring that all patient groups are represented fairly in the predictive models [118].

H. REAL-WORLD DEPLOYMENTS

BCFL has progressed beyond proof-of-concept demonstrations and now supports production-grade solutions in drug discovery, infectious disease triage, precision oncology, and ophthalmology. Table 6 summarises four flagship initiatives, illustrating that BC-FL can scale from single-GPU workstations to continent-wide hospital networks while satisfying HIPAA/GDPR audits, operating on non-IID data, and withstanding adversarial clients.

1) MELLODDY (DRUG DISCOVERY)

Ten pharmaceutical companies pooled bioactivity data on 68 proprietary compounds via a Quorum side-chain. Graph-neural QSAR models yielded a 26% improvement in external hit-rate over the best solo participant, with only 0.9 s additional latency per aggregation round [119].

2) FedRMD–COVID-CXR (DIAGNOSTICS)

In 40 simulated hospitals, Hyperledger Fabric using the FedRMD protocol improved chest-X-ray triage AUROC from 0.84 to 0.92, maintained 282 tps, and maintained block confirmation under 0.6 s during a 12,000-block stress test [120].

3) TNBC-FL (ONCOLOGY)

Six European oncology centres jointly analysed whole-slide images to predict response to neoadjuvant chemotherapy in triple-negative breast cancer. Collaborative training increased the prognostic C-index by 0.05 compared to local models; the open-source pipeline now ships with an optional Fabric ledger for on-chain gradient digest logging [121].

4) RETINA-BCFL (OPHTHALMOLOGY)

A Fabric-based federation of eight clinics in Singapore, China, and Taiwan trained on 27145 fundus/OCT images, attaining AUC = 0.970 and resisting label-flipping attacks while adding only 5s blockchain overhead per round [122].

5) ARCHITECTURAL INSIGHTS

Quorum's private-transaction infrastructure protects IP-sensitive drug-discovery information, and the MELLODDY and RAFT consensus minimizes latency. Hyperledger Fabric has enjoyed superiority in medical-imaging applications, thanks to its modular access control on top of channels and modular consensus. FedAvg, while still dominant, was outperformed by domain-specific aggregators like FedRMD's radiomics-aware metadata distillation and Retina-BCFL's attention-weighted scheme in improving performance under heterogeneous data quality.

Globally, out-of-order Extreme FL rounds were the main bottleneck for heavy upstream calculations (e.g., MELLODDY docking simulations). Bloat in the ledger was reduced by offloading to IPFS (Retina-BCFL), and incentive misalignment was solved by smart-contract-based Shapley assignments (TNBC-FL). All four systems delivered near-real-time performance within typical clinical service-level agreements, even in these challenging environments.

V. CHALLENGES AND FUTURE DIRECTIONS

Efficient and safe delivery of home care services, as well as intelligent health monitoring systems, pose a significant challenge to the healthcare industry. The challenges are due to a variety of factors, such as patient logistics, privacy, data handling, and compatibility with cutting-edge technologies, such as blockchain and FL. This section outlines the key challenges and proposes future research directions to address them.

A. ROBUSTNESS CHALLENGES OF FedAvg AND FedSGD

FedAvg and FedSGD are the primitives of federated learning but lack a sufficient amount of adversarial and Byzantine robustness. In particular, FedAvg's naive averaging across clients' updates renders it extremely vulnerable to

gradient poisoning and label-flipping attacks. Empirical evidence [123], [124] shows that under a worst-case 90% malicious clients, FedAvg's test accuracy can deteriorate as low as 0% on common benchmarks like MNIST and FMNIST. Likewise, FedSGD, though providing updates of higher rates, suffers from noise amplification, especially in Gaussian attacks, and results in accelerated loss of accuracy compared to more robust variants such as R-FedM.

1) BLOCKCHAIN-ENHANCED BYZANTINE ROBUSTNESS

To address these vulnerabilities, recent research has integrated blockchain-based technologies into FL systems. Blockchain provides decentralized trust, tamper-proofing, and reward-compatible mechanisms, which are necessary in withstanding adversarial behavior. For example, Block-BRFL [125] substitutes the central aggregator with smart contracts, thus removing single points of failure. With 30% Byzantine clients and non-IID data, it maintains over 80% accuracy. In addition, token-based incentive mechanisms [123] prevent aberrant behavior through the punishment of deviant contributions, while privacy-preserving incentives like homomorphic encryption [126] protect the integrity of FedSGD's updates without violating data privacy.

2) DEFENSIVE STRATEGIES AGAINST ADVERSARIAL ATTACKS

Several algorithmic methods have also been proposed to enhance FedAvg and FedSGD robustness. Robust aggregation methods like trimmed mean and coordinate-wise median perform much better than naive averaging under Gaussian noise and poisoning attacks [124]. Dynamic filtering methods like Grubbs test-based outlier rejection enhance performance in non-IID distributions by rejecting hateful updates strategically. Attention-guided feature-based defenses, e.g., the Federated Defense with Biased Attention (FDBA) approach [127], utilize attention mechanisms to separate and reduce priority to adversarial gradients, and detection rates of up to 92% in federated LLM training setups.

3) BLOCKCHAIN-AUGMENTED DEFENSE MECHANISMS

Beyond algorithmic methods, blockchain offers other resilience in the form of immutable logging and consensus-verified validation. Methods such as BRFL [81] hash model updates and commit them on-chain to prevent tampering with them, and consensus algorithms such as PBFT discard suspicious updates prior to aggregation [81]. These measures enhance federated trust and auditability, especially if there are malicious participants.

B. PATIENT TRANSFER AND DATA PRIVACY

Transferring patients from their homes to hospitals for routine checkups often causes stress and increases the risk of exposure to illnesses. Although home healthcare services aim to mitigate these issues, they face significant hurdles in ensuring the privacy and security of patient data. Blockchain technology offers a potential solution by providing an open and responsible data protection method. However, optimization is still required to balance the privacy, security, and operational efficiency [128].

C. COMPUTATIONAL OVERHEADS

Implementing blockchain-enabled FL systems incurs substantial computational costs, particularly owing to encryption methods that enhance privacy but increase the processing overhead. Although techniques such as gradient compression can reduce these costs, they may compromise the quality of the global model. Efficient management of these trade-offs is crucial for the practical deployment of such systems [129].

D. GAS CONSUMPTION IN HOMOMORPHIC ENCRYPTION (HE)-BASED SYSTEMS

Homomorphic encryption maintains gradient privacy through the allowance of computation on ciphertexts, but at the expense of extremely high gas overhead at the level of smart-contracts. Each ciphertext is tens of kilobytes in size and consists of tens of high-gas `SSTORE`, `CALLDATA`, and modular-arithmetic big-integer opcodes. Therefore, end-to-end training iterations can be uneconomical on public chains that pay gas dynamically [130].

1) COST DETERMINANTS

Gas consumption increases with (i) ciphertext bit-size, (ii) multiplicative circuit depth, and (iii) frequency of aggregation calls on-chain. Use even of a single 2,048-bit BFV ciphertext 5 kB may exceed 100,000 gas, e.g., with ensuing on-chain modular-multiplication of two such ciphertexts, even doubling the same. Unpredictable base-fees only add to the cost unpredictability, rendering predictable funding of longitudinal clinical studies difficult [130].

E. SMART CONTRACT VULNERABILITIES

The faulty implementation of smart contracts can lead to security and privacy vulnerabilities, compromising the integrity of BCFL models. Addressing these vulnerabilities requires thorough security audits and robust design practices to ensure reliable execution and data management within smart contracts [131].

F. HETEROGENEOUS DEVICE CAPABILITIES

Devices in federated networks often have varied storage, computing, and communication capacities, which leads to inconsistencies in data processing and transmission. Ensuring fault tolerance and resilience in diverse environments is critical for the successful implementation of BCFL systems [132].

G. COMMUNICATION BOTTLENECKS

Effective communication is a critical bottleneck in federated networks, especially when integrating various healthcare systems with non-standardized protocols. Efficient communication algorithms are necessary to minimize the communication load and enhance overall system performance [133].

H. SCALABILITY AND EFFICIENCY

1) THROUGHPUT AND LATENCY

Public blockchains (Layer 1), like Ethereum, sequentially process $\mathcal{O}(10)$–$\mathcal{O}(10^2)$ tps. In contrast, a single FL round with 100 hospitals can yield approximately 10^4 gradient-commit transactions. Even high-performance permissioned blockchains, like Hyperledger Fabric v2.5, reach a maximum of about 3,200 tps in a 16-peer deployment, with confirmation time increasing beyond 4 second [134], [135]. Its improved variant, **ACHealthChain**, provides 19.7% more throughput and 87% less latency using hierarchical ordering and gRPC batching. It slows down, though, because there are more than 32 peers online [136].

2) STATE GROWTH AND STORAGE

The blockchain ledger increases linearly per FL round because each model update contributes approximately 1–4 kilo byte of metadata. Unpruned, a weekly-round deployment of 2 years with 512 nodes would take up to 6.1 tera byte of SSD storage space. To prevent loss of this, model tensors are offloaded (e.g., on IPFS or S3) and retained on-chain only as a 256-bit SHA-2 digest. Storage needs decrease by more than 99%, but mechanisms to ensure data availability increase.

3) CONSENSUS IN HETEROGENEOUS ENVIRONMENTS

Byzantine fault-tolerant (BFT) protocols are very secure but poorly scalable. Proof-of-authority (PoA) works better under regular loads. However, it may fail under denial-of-service (DoS) attacks. More recent designs, such as directed acyclic graphs (DAGs) and shard-fusion systems, achieve over 10,000 tps but may cause consistency errors across shards, which in turn may corrupt the global FL model [137]. Personalized FL on DAGs (PFL-DAG) reduces error by 10–80% for non-IID setups but still does not have a formal mechanism to avoid double-spending [138].

4) LAYER-2 AND ROLL-UP SOLUTIONS

Layer-2 roll-ups combine numerous gradient commits and submit a single compressed proof on Layer 1, cutting down the fees of transactions by approximately 90% without compromising the security of Layer 1 [139]. Problems are bundling proofs from various FL optimizers and conflict resolution in time for clinical use cases that need real-time response and feedback.

I. REGULATORY COMPLIANCE ACROSS JURISDICTIONS

1) HIPAA (UNITED STATES)

HIPAA Privacy Rule (45 CFR §164.502) allows disclosure of protected health information (PHI) for health care operations, as long as the integrity and auditability requirements from the Security Rule are met. Immutable blockchain records and smart contracts maintaining records of differential privacy (DP) budgets can meet §164.312(b) (audit controls) and §164.312(c) (data integrity), while facilitating forensic analysis [134], [140].

2) GDPR (EUROPE)

GDPR allows processing of personal data for scientific purposes by Articles 6(1)(f) and 9(2)(j). But the authorization is incompatible with Article 17, which creates the ''right to erasure,'' a right to which the immutability of blockchain systems is inherently anathema. These three principal technical solutions to this problem have arisen. The former involves the use of chameleon hashes or mutable pointers, where allowed data controllers can modify block headers without incurring any cost on collision resistance and data integrity to other clients [141], [142].

3) EU AI ACT AND GLOBAL AI GOVERNANCE

Since August 2024, the EU AI Act regards most clinical blockchain-FL systems as *high-risk*. Such systems will be subject to strict regulations, such as risk management plans, human intervention, and clear declaration of data use [143]. Blockchain-based ''AI passports,'' pilot-tested in the EuCanImage project, enable compliance, and DP budget tracking can be evidence for risk mitigation. Beyond the EU, the NIST AI Risk Management Framework (RMF) v1.0 provides a voluntary but well-matched guideline for responsible AI [144].

J. FUTURE RESEARCH DIRECTIONS

As BCFL grows into maturity, future research needs to shift away from abstract architectures and towards deployable, secure, and scalable systems designed for actual healthcare settings. Under this subsection, seven key research directions are presented that need to be approached with caution: privacy-preserving verification using zero-knowledge proofs, cryptographic performance and gas efficiency, adaptive orchestration for device diversity, differential privacy trade-off modeling, blockchain sharding for scaling, post-quantum security incorporation, and verifiable model aggregation.

1) PRIVACY-PRESERVING TECHNIQUES WITH ZKPs

The integration of ZKPs into BCFL architectures can significantly enhance data confidentiality by enabling verification of local model updates without exposing raw gradients or patient information. This paradigm reduces the need for trusted third-party validators and lowers the verification burden on-chain. Future research should explore lightweight ZKP frameworks that can operate under constrained resources and support aggregation proofs across heterogeneous federated optimizers [145].

2) EFFICIENT CRYPTOGRAPHY AND GAS OPTIMIZATION

Homomorphic encryption and SMPC are good options to protect model training, but their extension towards smart contracts is still computationally costly and gas-intensive. The high gas price is a limiting factor for real-world deployments of HE-based BCFL on public chains. Therefore, there is a need to design cost-effective cryptographic protocols and push on-chain encryption operations to make it cost-effective

and ensure economic scalability [72]. Systematic security audits of smart contracts are also necessary to remove vulnerabilities and uphold system integrity.

3) ADAPTIVE STRATEGIES FOR DEVICE HETEROGENEITY

Federated learning systems running between various healthcare providers need to deal with heterogeneous device capabilities and connection statuses. Future research needs to develop adaptive orchestration techniques that adapt training loads, privacy budgets, and update rates according to device capacities. These solutions will enhance the system's robustness and promote fair contribution from edge devices like wearables and mobile sensors [58].

4) FINE-GRAINED DIFFERENTIAL PRIVACY (DP) TRADE-OFFS

Although differential privacy (DP) is pivotal in HIPAA and GDPR regulation compliance, strict DP parameters come at a cost of model performance. For example, $\varepsilon < 2$ on cancer data can decrease diagnostic accuracy by as much as 3%. Upcoming work should parameterize privacy–utility trade-offs for clinical data types (e.g., radiology, ICU, genomics), create adaptive noise schedulers that discretely adjust ε for under-represented classes, and quantify the energy cost of DP mechanisms on edge hardware [146], [147].

5) BLOCKCHAIN SHARDING FOR SCALABILITY

Monolithic blockchain ledgers suffer from throughput bottlenecks as the number of participating hospitals increases. Sharding, a technique that partitions the ledger into parallel chains, offers a scalable alternative. A recent prototype on Hyperledger Fabric for EHR-FL reduced validation latency by 58%. However, open challenges remain, including the design of efficient shard-handoff protocols during patient transfers and ensuring verifiability of model updates across shards [148]. The permissioned vs. permissionless blockchain distinction impacts FL performance greatly, particularly in healthcare. Permissioned blockchains enable high-speed consensus, reduced power usage, and access control by role, enabling support for GDPR-compliant, real-time FL aggregation. However, if trust is centralized, their reduced decentralization may pose a problem. Conversely, permissioned blockchains such as Ethereum are more transparent and auditable but plagued by poor throughput, wasteful energy consumption, and latency problems that compromise convergence in models and present privacy threats unless corrected through cryptography.

6) POST-QUANTUM SECURITY FOR LONG-TERM INTEGRITY

Post-quantum cryptography becomes a focal topic for any conventional cryptographic schemes like ECDSA due to quantum computing. BCFL systems must be pre-emptively advancing with post-quantum cryptography in order to ensure long-term reliability. The PQS-BFL protocol demonstrates the application of ML-DSA-65 in model signing with zero latency [149]. Directions towards the future involve lattice-based threshold signature schemes allowing secure global model approval in blockchain shards.

7) ZKPS FOR VERIFIABLE MODEL AGGREGATION

Zero-knowledge–based aggregation protocols like *zkDFL* have been demonstrated to be capable of certifying client updates with their existence without exposing sensitive information. This not only enhances transparency but also minimizes up to 40% of gas usage compared to naive on-chain storage [150]. Merging ZKPs with post-quantum signature schemes has the potential to facilitate both confidentiality and cryptographic resistance in future BCFL implementations.

VI. CONCLUSION

FL has emerged as a promising technology in the field of smart healthcare because it offers privacy-enhancing solutions and scalable networks for managing sensitive medical data. This study presents a comprehensive survey of FL applications in healthcare and discusses its foundational concepts, motivations, and technical requirements. By addressing the growing demand for data privacy, particularly in areas such as EHR management and remote health monitoring, FL provides an innovative approach to balancing patient confidentiality with the need for collaborative data analysis. Looking ahead, FL in smart healthcare is poised to mature rapidly, driving a shift from centralized to distributed healthcare data analytics. As healthcare technology continues to evolve, the integration of blockchain-enabled FL is expected to significantly enhance not only data privacy but also the operational efficiency and predictive accuracy of medical models. This advancement holds the potential to facilitate more personalized, timely, and effective patient care. Ongoing research into blockchain-enabled FL will likely open new avenues for secure, large-scale collaboration in precision medicine, enabling healthcare providers to jointly analyze distributed datasets without compromising sensitive patient information. The ability to decentralize control over healthcare data while maintaining robust security, privacy, and regulatory compliance will be a cornerstone in realizing the full potential of these emerging technologies. Furthermore, this survey highlighted critical future research directions for advancing BCFL, such as enhancing system scalability, privacy-preserving techniques, and interoperable frameworks for diverse healthcare data. Addressing these areas will help researchers and practitioners optimize these technologies to meet real-world healthcare needs.

REFERENCES

[1] Y. Zhao, Y. Qu, Y. Xiang, F. Chen, and L. Gao, "Context-aware consensus algorithm for blockchain-empowered federated learning," *IEEE Trans. Cloud Comput.*, vol. 12, no. 2, pp. 491–503, Apr. 2024, doi: 10.1109/TCC.2024.3372814.

[2] K. M. Sameera, S. Nicolazzo, M. Arazzi, A. Nocera, R. K. A. Rafidha, P. Vinod, and M. Conti, "Privacy-preserving in blockchain-based federated learning systems," *Comput. Commun.*, vol. 222, pp. 38–67, Jun. 2024, doi: 10.1016/j.comcom.2024.04.024.

[3] W. Issa, N. Moustafa, B. Turnbull, N. Sohrabi, and Z. Tari, "Blockchain-based federated learning for securing Internet of Things: A comprehensive survey," *ACM Comput. Surv.*, vol. 55, no. 9, p. 191, Sep. 2023, doi: 10.1145/3560816.

[4] M. M. Salim, L. T. Yang, and J. H. Park, "Privacy-preserving and scalable federated blockchain scheme for healthcare 4.0," *Comput. Netw.*, vol. 247, Jun. 2024, Art. no. 110472, doi: 10.1016/j.comnet.2024.110472.

[5] R. Fotohi, F. Shams Aliee, and B. Farahani, "Decentralized and robust privacy-preserving model using blockchain-enabled federated deep learning in intelligent enterprises," *Appl. Soft Comput.*, vol. 161, Aug. 2024, Art. no. 111764, doi: 10.1016/j.asoc.2024.111764.

[6] A. Afaq, Z. Ahmed, N. Haider, and M. Imran, "Blockchain-based collaborated federated learning for improved security, privacy and reliability," 2022, *arXiv:2201.08551*.

[7] J. Zhu, J. Cao, D. Saxena, S. Jiang, and H. Ferradi, "Blockchain-empowered federated learning: Challenges, solutions, and future directions," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–31, Feb. 2023.

[8] Y. Shahsavari, O. A. Dambri, Y. Baseri, A. Senhaji Hafid, and D. Makrakis, "Integration of federated learning and blockchain in healthcare: A tutorial," 2024, *arXiv:2404.10092*.

[9] H. Zhu, H. Zhang, and Y. Jin, "From federated learning to federated neural architecture search: A survey," *Complex Intell. Syst.*, vol. 7, no. 2, pp. 639–657, Jan. 2021, doi: 10.1007/s40747-020-00247-z.

[10] L. Cui, Y. Li, Y. Zhou, Y. Qu, and J. Liu, "Accelerating blockchain-enabled federated learning with clustered clients," *IEEE Trans. Big Data*, pp. 1–14, May 2024, doi: 10.1109/TBDATA.2024.3403390.

[11] R. Jin, J. Hu, G. Min, and J. Mills, "Lightweight blockchain-empowered secure and efficient federated edge learning," *IEEE Trans. Comput.*, vol. 72, no. 11, pp. 3314–3325, Nov. 2023, doi: 10.1109/TC.2023.3293731.

[12] C. Jain and P. Chaudhari, "Blockchain-aided privacy preserving framework for federated learning," in *Proc. Int. Conf. Intell. Innov. Technol. Comput., Electr. Electron. (IITCEE)*, Jan. 2024, pp. 1–6, doi: 10.1109/iitcee59897.2024.10467294.

[13] M. Ali, F. Naeem, M. Tariq, and G. Kaddoum, "Federated learning for privacy preservation in smart healthcare systems: A comprehensive survey," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 2, pp. 778–789, Feb. 2023.

[14] U. Zukaib, X. Cui, M. Hassan, S. Harris, H. J. Hadi, and C. Zheng, "Blockchain and machine learning in EHR security: A systematic review," *IEEE Access*, vol. 11, pp. 130230–130256, 2023, doi: 10.1109/ACCESS.2023.3333229.

[15] H. Li, C. Li, J. Wang, A. Yang, Z. Ma, Z. Zhang, and D. Hua, "Review on security of federated learning and its application in healthcare," *Future Gener. Comput. Syst.*, vol. 144, pp. 271–290, Jul. 2023, doi: 10.1016/j.future.2023.02.021.

[16] R. Myrzashova, S. H. Alsamhi, A. V. Shvetsov, A. Hawbani, and X. Wei, "Blockchain meets federated learning in healthcare: A systematic review with challenges and opportunities," *IEEE Internet Things J.*, vol. 10, no. 16, pp. 14418–14437, Aug. 2023, doi: 10.1109/JIOT.2023.3263598.

[17] M. Hiwale, R. Walambe, V. Potdar, and K. Kotecha, "A systematic review of privacy-preserving methods deployed with blockchain and federated learning for the telemedicine," *Healthcare Anal.*, vol. 3, Nov. 2023, Art. no. 100192, doi: 10.1016/j.health.2023.100192.

[18] M. J. Page et al., "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *PLOS Med.*, vol. 18, no. 3, Mar. 2021, Art. no. e1003583, doi: 10.1371/journal.pmed.1003583.

[19] Y. Jia, L. Xiong, Y. Fan, W. Liang, N. Xiong, and F. Xiao, "Blockchain-based privacy-preserving multi-tasks federated learning framework," *Connection Sci.*, vol. 36, no. 1, pp. 2299103–2299112, Jan. 2024, doi: 10.1080/09540091.2023.2299103.

[20] R. Yang, T. Zhao, F. R. Yu, M. Li, D. Zhang, and X. Zhao, "Blockchain-based federated learning with enhanced privacy and security using homomorphic encryption and reputation," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 21674–21688, Jun. 2024, doi: 10.1109/JIOT.2024.3379395.

[21] R. Cheng, Y. Sun, Y. Liu, L. Xia, D. Feng, and M. A. Imran, "Blockchain-empowered federated learning approach for an intelligent and reliable D2D caching scheme," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 7879–7890, Jun. 2022, doi: 10.1109/JIOT.2021.3103107.

[22] Z. Abou El Houda, H. Moudoud, B. Brik, and L. Khoukhi, "Blockchain-enabled federated learning for enhanced collaborative intrusion detection in vehicular edge computing," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 7661–7672, Jul. 2024, doi: 10.1109/TITS.2024.3351699.

[23] E. Goh, D.-Y. Kim, K. Lee, S. Oh, J.-E. Chae, and D.-Y. Kim, "Blockchain-enabled federated learning: A reference architecture design, implementation, and verification," *IEEE Access*, vol. 11, pp. 145747–145762, 2023, doi: 10.1109/ACCESS.2023.3345360.

[24] B. Annappa, S. Hegde, C. S. Abhijit, and S. Ambesange, "FedCure: A heterogeneity-aware personalized federated learning framework for intelligent healthcare applications in IoMT environments," *IEEE Access*, vol. 12, pp. 15867–15883, 2024, doi: 10.1109/ACCESS.2024.3357514.

[25] M. Al Asqah and T. Moulahi, "Federated learning and blockchain integration for privacy protection in the Internet of Things: Challenges and solutions," *Future Internet*, vol. 15, no. 6, p. 203, May 2023, doi: 10.3390/fi15060203.

[26] D. C. Nguyen, Q.-V. Pham, P. N. Pathirana, M. Ding, A. Seneviratne, Z. Lin, O. Dobre, and W.-J. Hwang, "Federated learning for smart healthcare: A survey," *ACM Comput. Surv.*, vol. 55, no. 3, pp. 1–37, Mar. 2023, doi: 10.1145/3501296.

[27] J. Qiu, L. Li, J. Sun, J. Peng, P. Shi, R. Zhang, Y. Dong, K. Lam, F. P.-W. Lo, B. Xiao, W. Yuan, N. Wang, D. Xu, and B. Lo, "Large AI models in health informatics: Applications, challenges, and the future," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 12, pp. 6074–6087, Dec. 2023, doi: 10.1109/JBHI.2023.3316750.

[28] M. Wang, T. Zhu, X. Zuo, D. Ye, S. Yu, and W. Zhou, "Public and private blockchain infusion: A novel approach to federated learning," *IEEE Internet Things J.*, vol. 11, no. 10, pp. 17525–17537, May 2024, doi: 10.1109/JIOT.2024.3360129.

[29] Y. Shanmugarasa, H.-Y. Paik, S. S. Kanhere, and L. Zhu, "A systematic review of federated learning from clients' perspective: Challenges and solutions," *Artif. Intell. Rev.*, vol. 56, no. S2, pp. 1773–1827, Nov. 2023, doi: 10.1007/s10462-023-10563-8.

[30] L. Liu, K. Fan, and M. Yang, "Federated learning: A deep learning model based on resnet18 dual path for lung nodule detection," *Multimedia Tools Appl.*, vol. 82, no. 11, pp. 17437–17450, May 2023, doi: 10.1007/s11042-022-14107-0.

[31] J. Wen, Z. Zhang, Y. Lan, Z. Cui, J. Cai, and W. Zhang, "A survey on federated learning: Challenges and applications," *Int. J. Mach. Learn. Cybern.*, vol. 14, no. 2, pp. 513–535, Feb. 2023, doi: 10.1007/s13042-022-01647-y.

[32] J. Liu, C. Chen, Y. Li, L. Sun, Y. Song, J. Zhou, B. Jing, and D. Dou, "Enhancing trust and privacy in distributed networks: A comprehensive survey on blockchain-based federated learning," *Knowl. Inf. Syst.*, vol. 66, no. 8, pp. 4377–4403, Aug. 2024, doi: 10.1007/s10115-024-02117-3.

[33] R. Myrzashova, S. H. Alsamhi, A. Hawbani, E. Curry, M. Guizani, and X. Wei, "Safeguarding patient data-sharing: Blockchain-enabled federated learning in medical diagnostics," *IEEE Trans. Sustain. Comput.*, vol. 10, no. 1, pp. 176–189, Jan. 2025, doi: 10.1109/TSUSC.2024.3409329.

[34] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst.*, vol. 2, 2020, pp. 429–450.

[35] M. Castro and B. Liskov, "Practical Byzantine fault tolerance," in *Proc. Symp. Operating Syst. Design Implement. (OSDI)*, vol. 99, no. 1999, pp. 173–186, 1999.

[36] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Aguera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, Fort Lauderdale, FL, USA, Apr. 2017, pp. 1273–1282.

[37] A. Ali, B. A. S. Al-Rimy, T. T. Tin, S. N. Altamimi, S. N. Qasem, and F. Saeed, "Empowering precision medicine: Unlocking revolutionary insights through blockchain-enabled federated learning and electronic medical records," *Sensors*, vol. 23, no. 17, p. 7476, Aug. 2023.

[38] X. Cheng, W. Tian, F. Shi, M. Zhao, S. Chen, and H. Wang, "A blockchain-empowered cluster-based federated learning model for blade icing estimation on IoT-enabled wind turbine," *IEEE Trans. Ind. Informat.*, vol. 18, no. 12, pp. 9184–9195, Dec. 2022, doi: 10.1109/TII.2022.3159684.

[39] Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Blockchain and federated learning for privacy-preserved data sharing in industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 16, no. 6, pp. 4177–4186, Jun. 2020, doi: 10.1109/TII.2019.2942190.

[40] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020, doi: 10.1109/MSP.2020.2975749.

[41] L. Lyu, H. Yu, X. Ma, C. Chen, L. Sun, J. Zhao, Q. Yang, and P. S. Yu, "Privacy and robustness in federated learning: Attacks and defenses," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 8726–8746, Jul. 2024, doi: 10.1109/TNNLS.2022.3216981.

[42] M. Chen, Y. Hao, K. Hwang, L. Wang, and L. Wang, "Disease prediction by machine learning over big data from healthcare communities," *IEEE Access*, vol. 5, pp. 8869–8879, 2017, doi: 10.1109/ACCESS.2017.2694446.

[43] Z. Tang, H.-S. Wong, and Z. Yu, "Privacy-preserving federated learning with domain adaptation for multi-disease ocular disease recognition," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 6, pp. 3219–3227, Jun. 2024, doi: 10.1109/JBHI.2023.3305685.

[44] D. Das, P. Chatterjee, S. Banerjee, U. Ghosh, and M. S. Al-Numay, "Blockchain-enabled federated learning for security and privacy in consumer electronics devices," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 2262–2270, Feb. 2025, doi: 10.1109/TCE.2025.3528934.

[45] B. Jia, X. Zhang, J. Liu, Y. Zhang, K. Huang, and Y. Liang, "Blockchain-enabled federated learning data protection aggregation scheme with differential privacy and homomorphic encryption in IIoT," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 4049–4058, Jun. 2022, doi: 10.1109/TII.2021.3085960.

[46] J. Klein Brinke, M. van der Linden, and P. Havinga, "Using federated learning and channel state information-based sensing for scalable and realistic at-home healthcare," in *Proc. Eur. Interdiscipl. Cybersecurity Conf.*, Xanthi, Greece, Jun. 2024, pp. 186–193, doi: 10.1145/3655693.3660254.

[47] S. M. Rajagopal, M. Supriya, and R. Buyya, "Blockchain integrated federated learning in edge/fog/cloud systems for IoT-based healthcare applications: A survey," in *Federated Learning: Principles, Paradigms, and Applications*, 1st ed. New York, NY, USA: Academic, 2024, p. 33.

[48] W. Ni, H. Ao, H. Tian, Y. C. Eldar, and D. Niyato, "FedSL: Federated split learning for collaborative healthcare analytics on resource-constrained wearable IoMT devices," *IEEE Internet Things J.*, vol. 11, no. 10, pp. 18934–18935, May 2024, doi: 10.1109/JIOT.2024.3370985.

[49] M. Jafari and F. Adibnia, "Securing IoMT healthcare systems with federated learning and BigchainDB," *Future Gener. Comput. Syst.*, vol. 165, Apr. 2025, Art. no. 107609, doi: 10.1016/j.future.2024.107609.

[50] S. H. Alsamhi, R. Myrzashova, A. Hawbani, S. Kumar, S. Srivastava, L. Zhao, X. Wei, M. Guizan, and E. Curry, "Federated learning meets blockchain in decentralized data sharing: Healthcare use case," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19602–19615, Jun. 2024, doi: 10.1109/JIOT.2024.3367249.

[51] B. Akoramurthy, B. Surendiran, K. Dhivya, S. Chowdhury, R. Govindaraj, A. Mehbodniya, and J. L. Webber, "Blockchain-based federated learning in Internet of Health Things," in *Federated Learning for Digital Healthcare Systems*, A. L. Imoize, M. S. Obaidat, and H. H. Song, Eds., New York, NY, USA: Academic, 2024, pp. 175–201, doi: 10.1016/B978-0-443-13897-3.00010-2.

[52] N. Waheed, A. Ur Rehman, A. Nehra, M. Farooq, N. Tariq, M. A. Jan, F. Khan, A. Z. Alalmaie, and P. Nanda, "FedBlockHealth: A synergistic approach to privacy and security in IoT-enabled healthcare through federated learning and blockchain," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2023, pp. 3855–3860, doi: 10.1109/globecom54140.2023.10437356.

[53] A. Fu, X. Zhang, N. Xiong, Y. Gao, H. Wang, and J. Zhang, "VFL: A verifiable federated learning with privacy-preserving for big data in industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 18, no. 5, pp. 3316–3326, May 2022, doi: 10.1109/TII.2020.3036166.

[54] S. Singh, S. Rathore, O. Alfarraj, A. Tolba, and B. Yoon, "A framework for privacy-preservation of IoT healthcare data using federated learning and blockchain technology," *Future Gener. Comput. Syst.*, vol. 129, pp. 380–388, Apr. 2022, doi: 10.1016/j.future.2021.11.028.

[55] C. Shiranthika, P. Saeedi, and I. V. Bajic, "Decentralized learning in healthcare: A review of emerging techniques," *IEEE Access*, vol. 11, pp. 54188–54209, 2023, doi: 10.1109/ACCESS.2023.3281832.

[56] G. Hegde, S. Bekal, P. D. Shenoy, and K. R. Venugopal, "Preserving privacy and security of electronic health records using blockchain-based federated learning (BFL) framework," in *Proc. IEEE 11th Region 10 Humanitarian Technol. Conf. (R10-HTC)*, Rajkot, India, Oct. 2023, pp. 853–859, doi: 10.1109/r10-htc57504.2023.10461823.

[57] S. Aich, N. K. Sinai, S. Kumar, M. Ali, Y. R. Choi, M.-I. Joo, and H.-C. Kim, "Protecting personal healthcare record using blockchain & federated learning technologies," in *Proc. 23rd Int. Conf. Adv. Commun. Technol. (ICACT)*, Feb. 2021, pp. 109–112, doi: 10.23919/ICACT51234.2021.9370566.

[58] S. Roy and D. Bera, "A blockchain-based verifiable aggregation for federated learning and secure sharing in healthcare," in *Proc. IEEE Int. Conf. Adv. Netw. Telecommun. Syst. (ANTS)*, Jaipur, India, Dec. 2023, pp. 165–170, doi: 10.1109/ants59832.2023.10469165.

[59] A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, "Collaborative federated learning for healthcare: Multi-modal COVID-19 diagnosis at the edge," *IEEE Open J. Comput. Soc.*, vol. 3, pp. 172–184, 2022, doi: 10.1109/OJCS.2022.3206407.

[60] A. P. Kalapaaking, I. Khalil, and X. Yi, "Blockchain-based federated learning with SMPC model verification against poisoning attack for healthcare systems," *IEEE Trans. Emerg. Topics Comput.*, vol. 12, no. 1, pp. 269–280, Jan./Mar. 2024, doi: 10.1109/TETC.2023.3268186.

[61] Z. Lian, W. Wang, Z. Han, and C. Su, "Blockchain-based personalized federated learning for Internet of Medical Things," *IEEE Trans. Sustain. Comput.*, vol. 8, no. 4, pp. 694–702, Oct./Dec. 2023, doi: 10.1109/TSUSC.2023.3279111.

[62] A. Lakhan, M. A. Mohammed, J. Nedoma, R. Martinek, P. Tiwari, A. Vidyarthi, A. Alkhayyat, and W. Wang, "Federated-learning based privacy preservation and fraud-enabled blockchain IoMT system for healthcare," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 2, pp. 664–672, Feb. 2023, doi: 10.1109/JBHI.2022.3165945.

[63] J. A. Alzubi, O. A. Alzubi, A. Singh, and M. Ramachandran, "Cloud-IIoT-based electronic health record privacy-preserving by CNN and blockchain-enabled federated learning," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 1080–1087, Jan. 2023, doi: 10.1109/TII.2022.3189170.

[64] R. Kumar, A. A. Khan, J. Kumar, N. A. Golilarz, S. Zhang, Y. Ting, C. Zheng, and W. Wang, "Blockchain-federated-learning and deep learning models for COVID-19 detection using CT imaging," *IEEE Sensors J.*, vol. 21, no. 14, pp. 16301–16314, Jul. 2021, doi: 10.1109/JSEN.2021.3076767.

[65] M. Guduri, C. Chakraborty, U. Maheswari, and M. Margala, "Blockchain-based federated learning technique for privacy preservation and security of smart electronic health records," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2608–2617, Feb. 2024, doi: 10.1109/TCE.2023.3315415.

[66] J. Passerat-Palmbach, T. Farnan, M. McCoy, J. D. Harris, S. T. Manion, H. L. Flannery, and B. Gleim, "Blockchain-orchestrated machine learning for privacy preserving federated learning in electronic health data," in *Proc. IEEE Int. Conf. Blockchain (Blockchain)*, Nov. 2020, pp. 550–555, doi: 10.1109/Blockchain50366.2020.00080.

[67] Z. Abou El Houda, A. S. Hafid, L. Khoukhi, and B. Brik, "When collaborative federated learning meets blockchain to preserve privacy in healthcare," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 2455–2465, Sep./Oct. 2023, doi: 10.1109/TNSE.2022.3211192.

[68] H. Elayan, M. Aloqaily, and M. Guizani, "Deep federated learning for IoT-based decentralized healthcare systems," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, Jun. 2021, pp. 105–109, doi: 10.1109/IWCMC51323.2021.9498820.

[69] M. J. Baucas, P. Spachos, and K. N. Plataniotis, "Federated learning and blockchain-enabled fog-IoT platform for wearables in predictive healthcare," *IEEE Trans. Computat. Social Syst.*, vol. 10, no. 4, pp. 1732–1741, Aug. 2023, doi: 10.1109/TCSS.2023.3235950.

[70] R. Durga and E. Poovammal, "FLED-block: Federated learning ensembled deep learning blockchain model for COVID-19 prediction," *Frontiers Public Health*, vol. 10, Jun. 17, 2022, Art. no. 892499, doi: 10.3389/fpubh.2022.892499.

[71] V. A. Patel, P. Bhattacharya, S. Tanwar, R. Gupta, G. Sharma, P. N. Bokoro, and R. Sharma, "Adoption of federated learning for healthcare informatics: Emerging applications and future directions," *IEEE Access*, vol. 10, pp. 90792–90826, 2022, doi: 10.1109/ACCESS.2022.3201876.

[72] E. Ashraf, N. F. F. Areed, H. Salem, E. H. Abdelhay, and A. Farouk, "FIDChain: Federated intrusion detection system for blockchain-enabled IoT healthcare applications," *Healthcare*, vol. 10, no. 6, p. 1110, Jun. 2022, doi: 10.3390/healthcare10061110.

[73] Y. Liu, W. Yu, Z. Ai, G. Xu, L. Zhao, and Z. Tian, "A blockchain-empowered federated learning in healthcare-based cyber physical systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 5, pp. 2685–2696, Sep./Oct. 2022, doi: 10.1109/TNSE.2022.3168025.

[74] H. Jin, X. Dai, J. Xiao, B. Li, H. Li, and Y. Zhang, "Cross-cluster federated learning and blockchain for Internet of Medical Things," *IEEE Internet Things J.*, vol. 8, no. 21, pp. 15776–15784, Nov. 2021, doi: 10.1109/JIOT.2021.3081578.

[75] W. Moulahi, I. Jdey, T. Moulahi, M. Alawida, and A. Alabdulatif, "A blockchain-based federated learning mechanism for privacy preservation of healthcare IoT data," *Comput. Biol. Med.*, vol. 167, Dec. 2023, Art. no. 107630, doi: 10.1016/j.compbiomed.2023.107630.

[76] A. Rehman, S. Abbas, M. A. Khan, T. M. Ghazal, K. M. Adnan, and A. Mosavi, "A secure healthcare 5.0 system based on blockchain technology entangled with federated learning technique," *Comput. Biol. Med.*, vol. 150, Nov. 2022, Art. no. 106019, doi: 10.1016/j.compbiomed.2022.106019.

[77] M. A. Rahman, M. S. Hossain, M. S. Islam, N. A. Alrajeh, and G. Muhammad, "Secure and provenance enhanced Internet of Health Things framework: A blockchain managed federated learning approach," *IEEE Access*, vol. 8, pp. 205071–205087, 2020, doi: 10.1109/ACCESS.2020.3037474.

[78] Z. Zhang, X. Cao, J. Jia, and N. Z. Gong, "FLDetector: Defending federated learning against model poisoning attacks via detecting malicious clients," in *Proc. KDD*, 2022, pp. 2545–2555.

[79] Y. Ren, M. Hu, Z. Yang, G. Feng, and X. Zhang, "BPFL: Blockchain-based privacy-preserving federated learning against poisoning attack," *Inf. Sci.*, vol. 665, Apr. 2024, Art. no. 120377, doi: 10.1016/j.ins.2024.120377.

[80] Y. Qu, M. P. Uddin, C. Gan, Y. Xiang, L. Gao, and J. Yearwood, "Blockchain-enabled federated learning: A survey," *ACM Comput. Surv.*, vol. 55, no. 4, pp. 1–35, Nov. 2022, doi: 10.1145/3524104.

[81] Y. Li, C. Xia, C. Li, and T. Wang, "BRFL: A blockchain-based Byzantine-robust federated learning model," *J. Parallel Distrib. Comput.*, vol. 196, Feb. 2025, Art. no. 104995, doi: 10.1016/j.jpdc.2024.104995.

[82] Y. Cao, D. Liu, S. Zhang, T. Wu, F. Xue, and H. Tang, "T-FedHA: A trusted hierarchical asynchronous federated learning framework for Internet of Things," *Expert Syst. Appl.*, vol. 245, Jul. 2024, Art. no. 123006, doi: 10.1016/j.eswa.2023.123006.

[83] F. Wilhelmi, N. Afraz, E. Guerra, and P. Dini, "The implications of decentralization in blockchained federated learning: Evaluating the impact of model staleness and inconsistencies," *Comput. Netw.*, vol. 245, May 2024, Art. no. 110361, doi: 10.1016/j.comnet.2024.110361.

[84] H. Zhang, S. Jiang, and S. Xuan, "Decentralized federated learning based on blockchain: Concepts, framework, and challenges," *Comput. Commun.*, vol. 216, pp. 140–150, Feb. 2024, doi: 10.1016/j.comcom.2023.12.042.

[85] H. Cai, L. Gao, J. Wang, and F. Li, "Reliable incentive mechanism in hierarchical federated learning based on two-way reputation and contract theory," *Future Gener. Comput. Syst.*, vol. 159, pp. 533–544, Oct. 2024, doi: 10.1016/j.future.2024.05.045.

[86] J. Senanayake, H. Kalutarage, A. Petrovski, L. Piras, and M. O. Al-Kadri, "Defendroid: Real-time Android code vulnerability detection via blockchain federated neural network with XAI," *J. Inf. Secur. Appl.*, vol. 82, May 2024, Art. no. 103741, doi: 10.1016/j.jisa.2024.103741.

[87] B. Chhetri, S. Gopali, R. Olapojoye, S. Dehbashi, and A. S. Namin, "A survey on blockchain-based federated learning and data privacy," in *Proc. IEEE 47th Annu. Comput., Softw., Appl. Conf. (COMPSAC)*, Jun. 2023, pp. 1311–1318, doi: 10.1109/COMPSAC57700.2023.00199.

[88] R. Lin, F. Wang, S. Luo, X. Wang, and M. Zukerman, "Time-efficient blockchain-based federated learning," *IEEE/ACM Trans. Netw.*, vol. 32, no. 6, pp. 4885–4900, Dec. 2024, doi: 10.1109/TNET.2024.3436862.

[89] Y. Tian, Z. Guo, J. Zhang, and Z. Al-Ars, "DFL: High-performance blockchain-based federated learning," 2021, *arXiv:2110.15457*.

[90] D. Commey, S. Hounsinou, and G. V. Crosby, "Securing health data on the blockchain: A differential privacy and federated learning framework," 2024, *arXiv:2405.11580*.

[91] D. Commey, S. G. Hounsinou, and G. V. Crosby, "Post-quantum secure blockchain-based federated learning framework for healthcare analytics," *IEEE Netw. Lett.*, vol. 7, no. 2, pp. 126–129, Jun. 2025, doi: 10.1109/LNET.2025.3563434.

[92] A. Mohey Eldin, E. Hossny, K. Wassif, and F. A. Omara, "Federated blockchain system (FBS) for the healthcare industry," *Sci. Rep.*, vol. 13, no. 1, p. 2569, Feb. 2023, doi: 10.1038/s41598-023-29813-4.

[93] L. Bai, H. Hu, Q. Ye, H. Li, L. Wang, and J. Xu, "Membership inference attacks and defenses in federated learning: A survey," 2024, *arXiv:2412.06157*.

[94] N. Dong, Z. Wang, J. Sun, M. Kampffmeyer, W. Knottenbelt, and E. Xing, "Defending against poisoning attacks in federated learning with blockchain," *IEEE Trans. Artif. Intell.*, vol. 5, no. 7, pp. 3743–3756, Jul. 2024. [Online]. Available: https://www.computer.org/csdl/journal/ai/2024/07/10471193/1VpY6lFFWko

[95] D. Makhija, X. Han, J. Ghosh, and Y. Kim, "Achieving fairness across local and global models in federated learning," 2024, *arXiv:2406.17102*.

[96] X. Wang and C. C. Yang, "Balancing fairness and performance in healthcare AI: A gradient reconciliation approach," 2025, *arXiv:2504.14388*.

[97] S. Shukla, A. Taneja, R. Bhushan, S. Shukla, and A. Arjaria, "Federated learning with differential privacy for breast cancer diagnosis enabling secure data sharing and model integrity," *Sci. Rep.*, vol. 15, p. 13061, Apr. 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-95858-2

[98] M. Firdaus, H. T. Larasati, and K. Hyune-Rhee, "Blockchain-based federated learning with homomorphic encryption for privacy-preserving healthcare data sharing," *Internet Things*, vol. 31, Mar. 2025, Art. no. 101579, doi: 10.1016/j.iot.2025.101579.

[99] H. Kaminaga, M. AbdelBaky, C. Fritz, J. Sonnekalb, and K. D. Bowers, "MPCFL: Towards multi-party computation for secure federated learning aggregation," in *Proc. UCC*, 2023, pp. 1–10, doi: 10.1145/3603166.3632144.

[100] J. Shao, F. Wu, and J. Zhang, "Selective knowledge sharing for privacy-preserving federated distillation without a good teacher," *Nature Commun.*, vol. 15, no. 1, p. 349, Jan. 2024. [Online]. Available: https://www.nature.com/articles/s41467-023-44383-9

[101] M. Gupta, M. Kumar, and Y. Gupta, "A blockchain-empowered federated learning-based framework for data privacy in lung disease detection system," *Comput. Hum. Behav.*, vol. 158, Sep. 2024, Art. no. 108302, doi: 10.1016/j.chb.2024.108302.

[102] J. Fu, N. Wang, and Y. Cai, "Privacy-preserving in healthcare blockchain systems based on lightweight message sharing," *Sensors*, vol. 20, no. 7, p. 1898, Mar. 2020, doi: 10.3390/s20071898.

[103] C. Gan, X. Xiao, Y. Zhang, Q. Zhu, J. Bi, D. K. Jain, and A. Saini, "An asynchronous federated learning-assisted data sharing method for medical blockchain," *Appl. Intell.*, vol. 55, p. 208, Dec. 2025, doi: 10.1007/s10489-024-06172-9.

[104] P. Qi, D. Chiaro, F. Giampaolo, and F. Piccialli, "A blockchain-based secure Internet of Medical Things framework for stress detection," *Inf. Sci.*, vol. 628, pp. 377–390, May 2023, doi: 10.1016/j.ins.2023.01.123.

[105] O. Aouedi, A. Sacco, K. Piamrat, and G. Marchetto, "Handling privacy-sensitive medical data with federated learning: Challenges and future directions," *IEEE J. Biomed. Health Informat.*, vol. 27, no. 2, pp. 790–803, Feb. 2023, doi: 10.1109/JBHI.2022.3185673.

[106] S. O. Hwang and A. Majeed, "Analysis of federated learning paradigm in medical domain: Taking COVID-19 as an application use case," *Appl. Sci.*, vol. 14, no. 10, p. 4100, May 2024, doi: 10.3390/app14104100.

[107] Z. Abbas, S. F. Ahmad, M. H. Syed, A. Anjum, and S. Rehman, "Exploring deep federated learning for the Internet of Things: A GDPR-compliant architecture," *IEEE Access*, vol. 12, pp. 10548–10574, 2024, doi: 10.1109/ACCESS.2023.3344029.

[108] A. Heidari, D. Javaheri, S. Toumaj, N. J. Navimipour, M. Rezaei, and M. Unal, "A new lung cancer detection method based on the chest CT images using federated learning and blockchain systems," *Artif. Intell. Med.*, vol. 141, Jul. 2023, Art. no. 102572, doi: 10.1016/j.artmed.2023.102572.

[109] K. Abbas, M. Afaq, T. Ahmed Khan, and W.-C. Song, "A blockchain and machine learning-based drug supply chain management and recommendation system for smart pharmaceutical industry," *Electronics*, vol. 9, no. 5, p. 852, May 2020, doi: 10.3390/electronics9050852.

[110] Y. Li, W. Liang, L. Peng, D. Zhang, C. Yang, and K.-C. Li, "Predicting drug-target interactions via dual-stream graph neural network," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 21, no. 4, pp. 948–958, Jul./Aug. 2024.

[111] Y. Chang, C. Fang, and W. Sun, "A blockchain-based federated learning method for smart healthcare," *Comput. Intell. Neurosci.*, vol. 2021, no. 1, Jan. 2021, Art. no. 4376418, doi: 10.1155/2021/4376418.

[112] A. Mishra, Y. Garg, O. J. Pandey, M. K. Shukla, A. V. Vasilakos, and R. M. Hegde, "A novel resource management framework for blockchain-based federated learning in IoT networks," *IEEE Trans. Sustain. Comput.*, vol. 9, no. 4, pp. 648–660, Jul. 2024, doi: 10.1109/TSUSC.2024.3358915.

[113] M. Li, F. R. Yu, P. Si, W. Wu, and Y. Zhang, "Resource optimization for delay-tolerant data in blockchain-enabled IoT with edge computing: A deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9399–9412, Oct. 2020, doi: 10.1109/JIOT.2020.3007869.

[114] T. R. S. Reddy and B. M. Beena, "AI integrated blockchain technology for secure health care—Consent-based secured federated transfer learning for predicting COVID-19 on Wearable Devices," in *Proc. Int. Conf. Innov. Comput. Commun.* Singapore: Springer, 2023, pp. 345–356, doi: 10.1007/978-981-19-2821-5_30.

[115] C. Dhasaratha, M. K. Hasan, S. Islam, S. Khapre, S. Abdullah, T. M. Ghazal, A. I. Alzahrani, N. Alalwan, N. Vo, and M. Akhtaruzzaman, "Data privacy model using blockchain reinforcement federated learning approach for scalable Internet of Medical Things," *CAAI Trans. Intell. Technol.*, vol. 12, pp. 123–145, Feb. 2024.

[116] A. Zahid and R. Sharma, "Personalized health care in a data-driven era: A post–COVID-19 retrospective," *Mayo Clinic Proc., Digit. Health*, vol. 1, no. 2, pp. 162–171, Jun. 2023.

[117] Y. K. Aluri, S. Katlagunta, T. Musunuru, A. Mylarapu, K. V. S. S. R. Kuruba, and J. V. Esther, "Healthcare supply chain security with ZeroKnowledge proof authentication in blockchain for personalized healthcare monitoring," in *Proc. 3rd Int. Conf. Distrib. Comput. Electr. Circuits Electron. (ICDCECE)*, Apr. 2024, pp. 1–6, doi: 10.1109/icdcece60827.2024.10549637.

[118] M. S. Alkatheiri and A. S. Alghamdi, "Blockchain-assisted cybersecurity for the Internet of Medical Things in the healthcare industry," *Electronics*, vol. 12, no. 8, p. 1801, Apr. 2023, doi: 10.3390/electronics12081801.

[119] W. Heyndrickx et al., "MELLODDY: Cross-pharma federated learning at unprecedented scale unlocks benefits in QSAR without compromising proprietary information," *J. Chem. Inf. Model.*, vol. 64, no. 7, pp. 2331–2344, 2023, doi: 10.1021/acs.jcim.3c00799.

[120] N. Li, R. Zhang, C. Zhu, W. Ou, W. Han, and Q. Zhang, "A data sharing method for remote medical system based on federated distillation learning and consortium blockchain," *Connection Sci.*, vol. 35, no. 1, Dec. 2023, Art. no. 2186315, doi: 10.1080/09540091.2023.2186315.

[121] J. Ogier du Terrail et al., "Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer," *Nature Med.*, vol. 29, no. 1, pp. 135–146, Jan. 2023, doi: 10.1038/s41591-022-02155-w.

[122] Z. L. Teo, X. Zhang, Y. Yang, L. Jin, C. Zhang, S. S. J. Poh, W. Yu, Y. Chen, J. B. Jonas, Y. X. Wang, W.-C. Wu, C.-C. Lai, Y. Liu, R. S. M. Goh, and D. S. W. Ting, "Privacy-preserving technology using federated learning and blockchain in protecting against adversarial attacks for retinal imaging," *Ophthalmology*, vol. 132, no. 4, pp. 484–494, Apr. 2025, doi: 10.1016/j.ophtha.2024.10.017.

[123] S. Zhao, J. Pu, X. Fu, L. Liu, and F. Dai, "Byzantine-robust federated learning with ensemble incentive mechanism," *Future Gener. Comput. Syst.*, vol. 159, pp. 272–283, Oct. 2024, doi: 10.1016/j.future.2024.05.017.

[124] W. Yao, B. Pan, Y. Hou, X. Li, and Y. Xia, "An adaptive model filtering algorithm based on grubbs test in federated learning," *Entropy*, vol. 25, no. 5, p. 715, Apr. 2023, doi: 10.3390/e25050715.

[125] W. Xiao, H. Zhong, Y. Dong, X. Tang, X. Zhang, and J. Han, "Block-BRFL: Blockchain-based high Byzantine robust federated learning method," in *Proc. 4th Int. Conf. Blockchain Technol. Inf. Secur. (ICBCTIS)*, Aug. 2024, pp. 96–105, doi: 10.1109/icbctis64495.2024.00023.

[126] Y. Miao, Z. Liu, H. Li, K. R. Choo, and R. H. Deng, "Privacy-preserving Byzantine-robust federated learning via blockchain systems," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 2848–2861, 2022, doi: 10.1109/TIFS.2022.3196274.

[127] C. Hu, Q. Hu, M. Zhang, and Z. Yang, "FDBA: Feature-guided defense against Byzantine and adaptive attacks in federated learning," *J. Inf. Secur. Appl.*, vol. 90, May 2025, Art. no. 104035, doi: 10.1016/j.jisa.2025.104035.

[128] Y. Zhang, Y. Liu, G. Yang, and J. Song, "ShareChain: Blockchain-enabled model for sharing patient data using federated learning and differential privacy," *Expert Syst.*, vol. 40, no. 5, 2023, Art. no. e13131, doi: 10.1111/exsy.13131.

[129] F. Yu, H. Lin, X. Wang, A. Yassine, and M. S. Hossain, "Blockchain-empowered secure federated learning system: Architecture and applications," *Comput. Commun.*, vol. 196, pp. 55–65, Dec. 2022, doi: 10.1016/j.comcom.2022.09.008.

[130] W. Yang, S. Wang, H. Cui, Z. Tang, and Y. Li, "A review of homomorphic encryption for privacy-preserving biometrics," *Sensors*, vol. 23, no. 7, p. 3566, Mar. 2023, doi: 10.3390/s23073566.

[131] Z. Cheng, Y. Liu, C. Wu, Y. Pan, L. Zhao, X. Deng, and C. Zhu, "Decentralized IoT data sharing: A blockchain-based federated learning approach with joint optimizations for efficiency and privacy," *Future Gener. Comput. Syst.*, vol. 160, pp. 547–563, Nov. 2024, doi: 10.1016/j.future.2024.06.035.

[132] S. Wang, M. Lee, S. Hosseinalipour, R. Morabito, M. Chiang, and C. G. Brinton, "Device sampling for heterogeneous federated learning: Theory, algorithms, and implementation," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, Vancouver, BC, Canada, May 2021, pp. 1–10, doi: 10.1109/INFOCOM42981.2021.9488906.

[133] M. S. Arbabi, C. Lal, N. R. Veeraragavan, D. Marijan, J. F. Nygård, and R. Vitenberg, "A survey on blockchain for healthcare: Challenges, benefits, and future directions," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 386–424, 1st Quart., 2023, doi: 10.1109/COMST.2022.3224644.

[134] L. Sadath, D. Mehrotra, and A. Kumar, "Scalability performance analysis of blockchain using hierarchical model in healthcare," *Blockchain Healthcare Today*, vol. 7, p. 30953, Apr. 2024, doi: 10.30953/bhty.v7.295.

[135] M. M. Khan, F. S. Khan, M. Nadeem, T. H. Khan, S. Haider, and D. Daas, "Scalability and efficiency analysis of hyperledger fabric and private Ethereum in smart contract execution," *Computers*, vol. 14, no. 4, p. 132, Apr. 2025.

[136] A. M. Tawfik, A. Al-Ahwal, A. S. T. Eldien, and H. H. Zayed, "ACHealthChain blockchain framework for access control and privacy preservation in healthcare," *Sci. Rep.*, vol. 15, no. 1, p. 16696, May 2025.

[137] J. Lee and W. Kim, "DAG-based blockchain sharding for secure federated learning with non-IID data," *Sensors*, vol. 22, no. 21, p. 8263, Oct. 2022, doi: 10.3390/s22218263.

[138] C. Huang, E. Liu, R. Wang, Y. Liu, H. Zhang, Y. Geng, J. Wang, and S. Han, "Personalized federated learning via directed acyclic graph based blockchain," *IET Blockchain*, vol. 4, no. 1, pp. 73–82, Mar. 2024.

[139] S. Ma and X. Zhang, "Integrating blockchain and ZK-ROLLUP for efficient healthcare data privacy protection system via IPFS," *Sci. Rep.*, vol. 14, no. 1, p. 11746, May 2024, doi: 10.1038/s41598-024-62292-9.

[140] M. M. Alam, M. S. Islam, S. Muntaha, M. T. Alam, and K. J. Tanha, "A secure and optimized healthcare data governance framework leveraging hyperledger fabric blockchain," in *Proc. 26th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Dec. 2023, pp. 1–6, doi: 10.1109/ICCIT60459.2023.10441320.

[141] S. Xu, J. Ning, X. Li, J. Yuan, X. Huang, and R. H. Deng, "A privacy-preserving and redactable healthcare blockchain system," *IEEE Trans. Services Comput.*, vol. 17, no. 2, pp. 364–377, Jan. 2024, doi: 10.1109/TSC.2024.3356595.

[142] A. Zafar, "Reconciling blockchain technology and data protection laws: Regulatory challenges, technical solutions, and practical pathways," *J. Cybersecur.*, vol. 11, no. 1, Jan. 2025, Art. no. tyaf002, doi: 10.1093/cybsec/tyaf002.

[143] D. Golpayegani, H. J. Pandit, and D. Lewis, "To be high-risk, or not to be—Semantic specifications and implications of the AI act's high-risk AI applications and harmonised standards," in *Proc. ACM Conf. Fairness, Accountability, Transparency (FAccT)*, 2023, pp. 905–915, doi: 10.1145/3593013.3594050.

[144] *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST, Gaithersburg, MD, USA, Jan. 2023, doi: 10.6028/NIST.AI.600-1.

[145] A. D. Dwivedi, R. Singh, U. Ghosh, R. Singh, D. Sinha, M. Alazab, and A. Jolfaei, "Privacy preserving authentication system based on non-interactive zero knowledge proof suitable for Internet of Things," *J. Ambient Intell. Humanized Comput.*, vol. 13, pp. 4639–4649, Sep. 2022, doi: 10.1007/s12652-021-03459-4.

[146] P. Gangwani, A. Perez-Pons, T. Bhardwaj, H. Upadhyay, S. Joshi, and L. Lagos, "Securing environmental IoT data using masked authentication messaging protocol in a DAG-based blockchain: IOTA tangle," *Future Internet*, vol. 13, no. 12, p. 312, Dec. 2021, doi: 10.3390/fi13120312.

[147] P. Gangwani, A. Perez-Pons, S. Joshi, H. Upadhyay, and L. Lagos, "Integration of data science and IoT with blockchain for Industry 4.0," in *Blockchain and its Applications in Industry 4.0* (Studies in Big Data), vol. 119, S. Namasudra and K. Akkaya, Eds., Singapore: Springer, 2023, pp. [insert page numbers if known], doi: 10.1007/978-981-19-8730-4_6.

[148] H. Dang, T. T. A. Dinh, D. Loghin, E.-C. Chang, Q. Lin, and B. C. Ooi, "Towards scaling blockchain systems via sharding," in *Proc. Int. Conf. Manage. Data*, Amsterdam, The Netherlands, Jun. 2019, pp. 123–140, doi: 10.1145/3299869.3319889.

[149] T. M. Fernández-Caramès and P. Fraga-Lamas, "Towards post-quantum blockchain: A review on blockchain cryptography resistant to quantum computing attacks," *IEEE Access*, vol. 8, pp. 21091–21116, 2020, doi: 10.1109/ACCESS.2020.2968985.

[150] M. Ahmadi and R. Nourmohammadi, "ZkFDL: An efficient and privacy-preserving decentralized federated learning with zero knowledge proof," in *Proc. IEEE 3rd Int. Conf. AI Cybersecur. (ICAIC)*, Feb. 2024, pp. 1–10, doi: 10.1109/icaic60265.2024.10433831.



NASIM NEZHADSISTANI (Graduate Student Member, IEEE) received the B.S. and M.Sc. degrees in electrical engineering from Isfahan University of Technology, Isfahan, Iran, in 2014 and 2017, respectively, where she is currently pursuing the Ph.D. degree in electrical engineering. Since 2022, she has been a Research Assistant with the Communication Systems Group (CSG), University of Zürich (UZH). Her research interests include blockchain technologies, distributed systems, machine learning, network security, and privacy-preserving mechanisms in peer-to-peer systems.



NAGHMEH S. MOAYEDIAN (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electrical engineering from Isfahan University of Technology, Isfahan, Iran, in 1999, 2002, and 2009, respectively. Since 2010, she has been with the ECE Department, Isfahan University of Technology. Her research interests include cloud networks, nanonetworks, and wireless networks.



BURKHARD STILLER (Member, IEEE) received the Dipl.-Inform. (M.Sc.) degree in computer science and the Ph.D. degree from the University of Karlsruhe, Germany, in 1990 and 1994, respectively. Since 2004, he has been with the Communication Systems Group (CSG), Department of Informatics (IfI), University of Zürich (UZH), Switzerland, as a Full Professor. He has published more than 300 articles, addressing decentralized systems with fully decentralized control, network and service management, the IoT, and telecommunication economics.